Laplace Transforms

This course aims to provide an introduction to *Laplace Transform* techniques – an alternative and powerful way of handling the differential equations that arise in a wide range of engineering applications.

1. INTRODUCTION

1.1. DIFFERENTIAL EQUATIONS

Differential equations are crucial for describing the behaviour of most engineering systems. In circuit theory, for example, the properties of idealized capacitors and inductors are described by the linear differential relationships

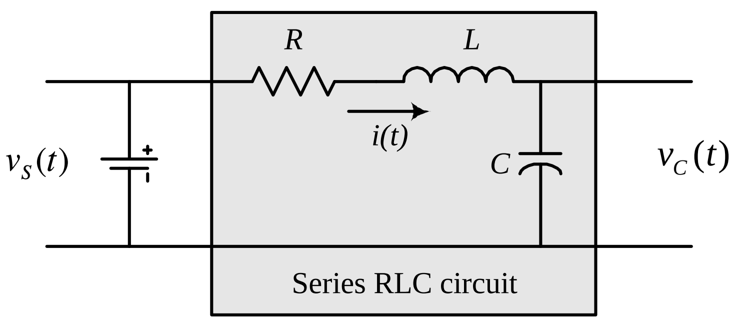
$$i = C\frac{dv_C}{dt} \qquad \text{and} \qquad v_L = L\frac{di}{dt} \tag{1}$$

so that networks of components will, in general, produce large and complicated sets of *coupled linear differential equations*, which relate the outputs to the inputs.

When one or more differential equations are required to model an engineering process, it is referred to as being a *dynamic system*.

Dynamic systems occur very commonly, not only in circuit applications, but also in control engineering, signal processing, music/acoustics, electromagnetics, solid-state devices, robotics and mechanics -- effectively in all situations where energy of some form is being stored, dissipated and/or propagated.

Example 1 :* *Series RLC circuit with Thévenin power source



The *L* and *C* energy storage elements within the RLC circuit are the source of its *dynamic response*, in the sense that their contribution is to provide the capability for the time-varying storage of energy - as indicated in equation (1) where v_L is the time-varying inductor voltage and likewise v_C is that for the capacitor.

Developing a differential equation for the entire system first requires the use of the physical laws relating to currents and voltages within a circuit, and then the use of the above ideal component relationships, leading to the equation

$$\frac{d^2 v_C}{dt^2} + \frac{R}{L}\frac{dv_C}{dt} + \frac{1}{LC} v_C = \frac{1}{LC} v_S(t) \qquad (2)$$

Here, the source voltage, $v_S(t)$, is the input to the circuit, and the solution of the differential equation, the capacitor voltage $v_C(t)$, is the output.

This is a particular type of differential equation - a *second-order, linear, differential equation with constant coefficients* ("constant" since it is assumed from the start that the properties of the circuit components do not change over time - hence *R*, *L* and *C* are just fixed numerical values). Since the equation is linear, the *superposition principle* holds, and it can be solved using the complementary function and particular integral approach:

1) First consider the homogeneous (free, undriven) differential equation

$$\frac{d^2 v_C}{dt^2} + \frac{R}{L}\frac{dv_C}{dt} + \frac{1}{LC} v_C = 0$$

and consider the *characteristic equation*

$$s^2 + \frac{R}{L}s + \frac{1}{LC} = 0$$

2) Solve this quadratic equation to get two roots s_1 and s_2 - these define the form of the *complementary function*:
(a) which, if s_1 and s_2 are real and distinct, takes the form

$$v_{C\mathbf{CF}} = A_1 \exp\left(s_1 t\right) + A_2 \exp\left(s_2 t\right)$$

(b) which if s_1 and s_2 are real and repeated, takes the form

$$v_{C\mathbf{CF}} = A_1 \exp\left(s_1 t\right) + A_2 t \exp\left(s_1 t\right)$$

(c) which if s_1 and s_2 are a complex conjugate pair ($= \sigma \pm j\omega$), takes the form

$$\begin{aligned} v_{C\mathbf{CF}} &= \exp\left(\sigma t\right)\left[A_1 \sin\left(\omega t\right) + A_2 \cos\left(\omega t\right)\right] \\ &= A \exp\left(\sigma t\right) \cos\left(\omega t + \phi\right) \end{aligned}$$

where the A_1 and A_2 (or *A* and ϕ) are *arbitrary constants* which can be determined later if *initial conditions* have been provided for the system.

For example, in the special case where $R = 3\Omega$, $L = 0.5\text{H}$ and $C = 0.25\text{F}$, we would have

$$\frac{d^2 v_C}{dt^2} + 6\frac{dv_C}{dt} + 8v_C = 0$$

and

$$\begin{aligned} & s^2 + 6s + 8 = 0 \\ \Rightarrow & \left(s+2\right)\left(s+4\right) = 0 \\ \Rightarrow & s = -2, \quad or \quad s = -4 \end{aligned}$$

so that the complementary function would be given by

$$v_{C\mathbf{CF}} = A_1 \exp\left(-2t\right) + A_2 \exp\left(-4t\right)$$

3) Then, consider the full inhomogeneous equation and establish an appropriate general form of trial *particular integral*, using the *method of undetermined coefficients*.

For example, if the driving input $v_S(t)$ is a unit ramp, then in the specific case above, the inhomogeneous equation is

$$\frac{d^2 v_C}{dt^2} + 6\frac{dv_C}{dt} + 8v_C = 8t$$

and an appropriate form of trial particular integral is the linear function

$$v_{C\mathbf{PI}} = K_1 + K_2 t$$

where K_1 and K_2 are the undetermined coefficients. To establish their values, differentiate the particular integral twice, substitute it in the equation, and compare the coefficients of the variables on the two sides:

$$6K_2 + 8\left(K_1 + K_2 t\right) = 8t$$
$$\Rightarrow 8K_2 = 8 \quad \text{and} \quad 6K_2 + 8K_1 = 0$$
$$\Rightarrow K_2 = 1 \quad \text{and} \quad K_1 = -0.75$$

The correct particular integral is hence

$$v_{C\mathbf{PI}} = -0.75 + t$$

4) Construct the general solution as the sum of the CF and PI:

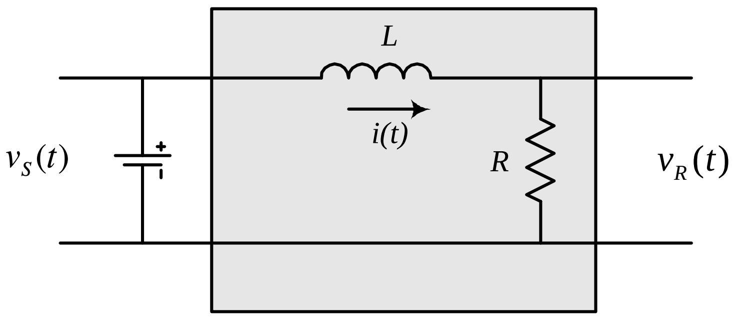
$$v_{C\mathbf{GS}} = v_{C\mathbf{CF}} + v_{C\mathbf{PI}} = A_1 \exp\left(-2t\right) + A_2 \exp\left(-4t\right) - 0.75 + t$$

5) If initial conditions are available, use them to establish numerical values for the arbitrary constants A_1 and A_2, giving a *particular solution*.

Note that the complementary function corresponds to the "free", "undriven", "transient", or "natural" behaviour of the system, while the particular integral represents the "driven", "forced" or "steady state" output.

Exactly the same approach can be used for first-order, linear, differential equation with constant coefficients.

Example 2 : ***LR circuit with Thévenin power source***



$v_S(t)$ is the driving input, and the resistor voltage is the output.

Consider the situation where the inductor is initially unfluxed, and the input has the particular form of a linear ramp, $v_S(t) = kt \quad \text{for } t \geq 0$ for some constant *k*.

In this example Kirchhoff's Voltage Law gives

$$v_S = L\frac{di}{dt} + Ri$$

and using the fact that $i = \frac{v_R}{R}$ it is possible to eliminate *i* to get

$$v_S = L\frac{d}{dt}\left(\frac{v_R}{R}\right) + R\left(\frac{v_R}{R}\right)$$

$$\Rightarrow \frac{L}{R}\frac{dv_R}{dt} + v_R = v_S$$

So that for the specified input, we need to solve the first-order, linear, differential equation

$$\frac{L}{R}\frac{dv_R}{dt} + v_R = kt$$

Just as for the second order case, the homogeneous equation is

$$\frac{L}{R}\frac{dv_R}{dt} + v_R = 0$$

and the characteristic equation is

$$\frac{sL}{R} + 1 = 0 \qquad\qquad \Rightarrow s = -\frac{R}{L}$$

So that the complementary function is

$$v_{R\mathbf{CF}} = A\exp\left(-\frac{Rt}{L}\right)$$

Next, as before, an appropriate form of trial particular integral is the linear function

$$v_{R\mathbf{PI}} = K_1 + K_2 t$$

where K_1 and K_2 are the undetermined coefficients. To establish their values, differentiate the particular integral, substitute it into the equation, and then compare the coefficients of the variables on the two sides:

$$\frac{L}{R}K_2 + \left(K_1 + K_2 t\right) = kt$$

$$\Rightarrow K_2 = k \quad \text{and} \quad K_1 = -\frac{kL}{R}$$

The correct particular integral is hence

$$v_{R\mathbf{PI}} = kt - \frac{kL}{R}$$

and the general solution is again the sum of the CF and PI:

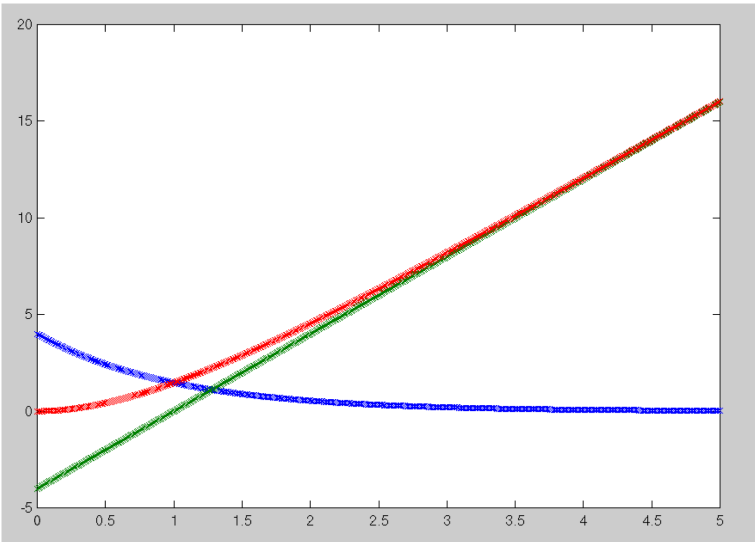
$$v_{R\mathbf{GS}} = \underbrace{A\exp\left(-{Rt}/{L}\right)}_{\text{CF}} + \underbrace{kt - \frac{kL}{R}}_{\text{PI}}$$

Here, we are also given the initial condition that the inductor is initially unfluxed - there is no energy storage and the circuit is initially in a state of *zero initial conditions* (ZIC).
The initial current is zero, and so $v_R\left(0\right) = 0$.

Hence, at $t = 0$, $v_R\left(0\right) = A - \frac{kL}{R} = 0$, so that $A = \frac{kL}{R}$ and the particular solution for the output voltage in this case is

$$v_R = \underbrace{\frac{kL}{R}\exp\left(-{Rt}/{L}\right)}_{\text{TRANSIENT}} + \underbrace{kt - \frac{kL}{R}}_{\text{STEADY STATE}} \qquad \text{for } t \geq 0$$

The plot below shows the typical form of the parts of the solution for some specific values of *R* and *L*.



$$v_R\left(t\right) = 4\exp\left(-t\right) + 4\left(t-1\right)$$

In engineering terms, systems such as those above - which can be described by ordinary linear differential equations with constant coefficients - are classed as *continuous linear time-invariant (LTI) systems*. Such systems arise very commonly and, rather than use differential equations to describe their properties, the Laplace Transform has been developed as a tool which simplifies their description and gives specific insights into their physical properties.

1.2. TRANSFORMS

Solving differential equations using *time-domain approaches*, such as that above, can be quite difficult, or even impossible, for more complicated equations..

An alternative is to transform the way of looking at the problem. For example, it is easier to think about walking four miles than it is to talk about a distance of 253,440 inches - such a linear change of scale is one of the simplest forms of transform.

Likewise, the equation of a unit circle centred at the origin is much simpler when viewed in polar coordinates ($r = 1$) than the same circle in Cartesian coordinates ($x^2 + y^2 = 1$) - switching between the two coordinate systems is a transform.

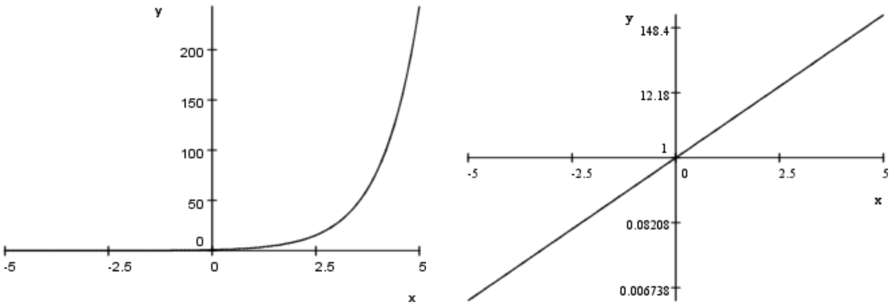
Taking a logarithm is a transform, designed to change a difficult operation (multiplication) into an easier one (addition).

$$\begin{tikzcd}[column sep=huge, row sep=huge] A, B \arrow[r, "\text{logarithm}"] \arrow[d, "\text{multiply}"'] & 10^a = A, 10^b = B \arrow[d, "\text{add exponents}"] \\ A \times B & 10^{a+b} \arrow[l, "\text{inverse logarithm}"'] \end{tikzcd}$$

Likewise the use of semi-log (linear-log) graph paper transforms the graph of an exponential function to that of a straight line:

$$y = a^x \quad \Rightarrow \quad \log_{10} y = \log_{10}\left(a^x\right) \quad \Rightarrow \log_{10} y = \log_{10}\left(a\right) x$$

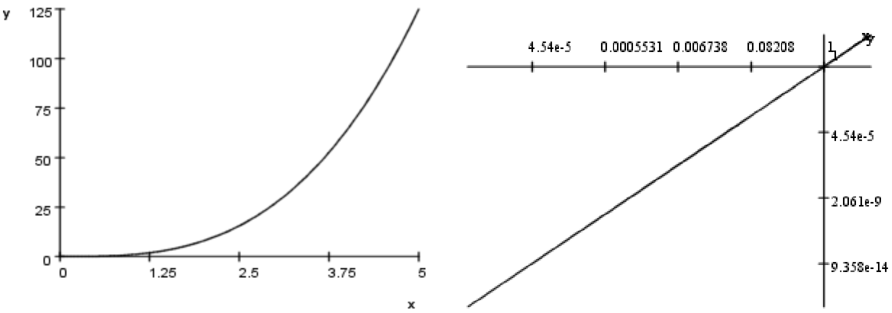
e.g. $y = 3^x$



Also, log-log paper transforms a power relationship to a straight line one:

$$y = x^n \quad \Rightarrow \quad \log_{10} y = \log_{10}\left(x^n\right) \quad \Rightarrow \log_{10} y = n\log_{10}\left(x\right)$$

e.g. $y = x^3$



This linearization effect is one of the reasons for the common use of *Bode plots* in engineering.

2. LAPLACE TRANSFORMS AND THEIR APPLICATIONS

The Laplace transform is designed specifically to simplify the handling and solution of differential equations, whether an individual equation, or large sets of coupled equations.

It is defined by the expression

$$F\left(s\right) = L\left[f\left(t\right)\right] \triangleq \int_0^{\infty} f\left(t\right)e^{-st}dt \tag{3}$$

Note:

1) It is assumed that $f\left(t\right) = 0 \quad \text{for} \quad t < 0$. Equation (3) is hence

known as the *single-sided Laplace transform*.

2) *s* is a complex variable - the transform changes the viewpoint from that of a (real) time-domain function $f(t)$ to that of a (complex) *s*-domain function $F(s)$.

3) $s = \sigma + j\omega$ is known as the *complex frequency variable* or the *Laplace variable*.

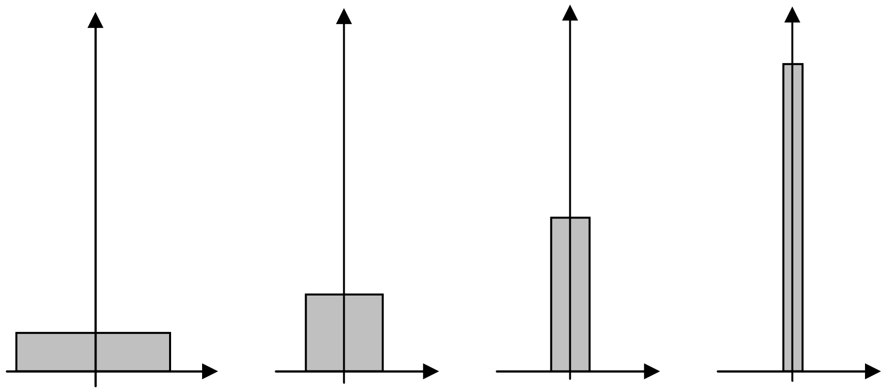
2.1. THE LAPLACE TRANSFORM OF SIMPLE SIGNALS

From the definition (3) it is possible to work out the *s*-domain versions of a range of common time-domain functions.

(1) The unit impulse (Dirac delta function)

Mathematical note: The delta function

The limit of a finite pulse of decreasing width but constant (unit) area gives one way of visualising the *Dirac delta function*:



which is defined by the two properties:

(1) $$\delta(t-\tau) = 0 \quad \text{when } t \neq \tau \tag{4}$$

(2) "sifting property" $$\int_{-\infty}^{\infty} f(t)\,\delta(t-\tau)\,dt = f(\tau) \tag{5}$$

Note that a consequence of the second property is that

$$\int_{-\infty}^{\infty} \delta(t-\tau)\,dt = 1 \tag{6}$$

This is a fictitious "ideal" impulse which is zero except for one (arbitrarily small) moment of time, but which has just enough (arbitrarily large) gain to ensure a total "integrated power" of unity.

Note that, in effect, multiplying any function by the delta function leads to:

$$f(t) \times \delta(t-\tau) = f(\tau)\delta(t-\tau) \tag{7}$$

so that it has the property of suppressing a function everywhere except at one moment in time.

From the definition of the Laplace Transform

$$L[\delta(t)] = \int_0^\infty \delta(t) e^{-st} dt = 1 \tag{8}$$

Hence $L[\delta(t)] = 1$.

Correspondingly, we refer to the *inverse Laplace transform*, and write:

$$L^{-1}[1] = \delta(t)$$

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(2) The unit step (Heavyside step function)

The unit step function is defined by the expression

$$\begin{aligned} u(t) &= 1 \quad t \geq 0 \\ &= 0 \quad t < 0 \end{aligned} \tag{9}$$

Hence

$$\begin{aligned} L[u(t)] &= \int_0^\infty u(t) e^{-st} dt \\ &= \int_0^\infty e^{-st} dt = \left[-\frac{1}{s} e^{-st}\right]_0^\infty \\ &= -\frac{1}{s}\left[e^{-st}\Big|_{t=\infty} - e^{-st}\Big|_{t=0}\right] = [0] - \left[-\frac{1}{s}\right] = \frac{1}{s} \end{aligned} \tag{10}$$

NOTE: Convergence...

Above, it has been assumed that $e^{-st}\Big|_{t=\infty} = 0$. But $s = \sigma + j\omega$ is a complex number - a complex variable in fact. Strictly, it is necessary to use *complex analysis*, *complex contour integration* and to consider *regions of convergence* when working with Laplace integrals and their inverses. Fortunately, in engineering practice, provided that we are working with positive-time functions, it turns out that these complexities can be avoided and it is safe to assume convergence. The Laplace transform of $u(t)$ is hence defined for all values of *s* except $s = 0$. **(Convergence issues are non-examinable in this course)**.

(3) The exponential function

If $f(t) = \exp(-at)$, then its Laplace transform is given by

$$\begin{aligned} L\left[e^{-at}\right] &= \int_0^\infty e^{-at}e^{-st}dt \\ &= \int_0^\infty e^{-(s+a)t}dt = \left[-\frac{1}{s+a}e^{-(s+a)t}\right]_0^\infty \\ &= -\frac{1}{s+a}\left[e^{-(s+a)t}\Big|_{t=\infty} - e^{-(s+a)t}\Big|_{t=0}\right] \end{aligned} \qquad (11)$$

Assuming that $e^{-(s+a)t}\Big|_{t=\infty} = 0$ gives

$$L\left[e^{-at}\right] = -\frac{1}{s+a}\big[[0]-[1]\big] = \frac{1}{s+a} \qquad (12)$$

NOTE - there is no restriction given here on the value of *a* - it can be positive, negative, or even a complex number.

(4) $\cos(\omega t)$

It is easy to confirm from the definition that the Laplace transform is a linear transform:

$$L\left[af(t) + bg(t)\right] = aF(s) + bG(s)$$

And since $\cos(\omega t)$ can be written in exponential form as

$$\cos\left(\omega t\right) = \frac{1}{2}\left(e^{j\omega t} + e^{-j\omega t}\right)$$

We can use the previous result to obtain

$$\begin{aligned} L\left[\cos\left(\omega t\right)\right] &= \frac{1}{2}\left(L\left[e^{j\omega t}\right] + L\left[e^{-j\omega t}\right]\right) \\ &= \frac{1}{2}\left(\frac{1}{s-j\omega} + \frac{1}{s+j\omega}\right) \\ &= \frac{1}{2}\left(\frac{\left(s+j\omega\right)+\left(s-j\omega\right)}{s^2+\omega^2}\right) = \frac{s}{s^2+\omega^2} \end{aligned} \tag{13}$$

Similarly, it's easy to show that $L\left[\sin\left(\omega t\right)\right] = \dfrac{\omega}{s^2+\omega^2}$.

We have hence derived the following five *Laplace Transform pairs*:

Time-domain	*s*-domain
$\delta\left(t\right)$	1
$u\left(t\right)$	$\frac{1}{s}$
$e^{-at}u\left(t\right)$	$\frac{1}{s+a}$
$\cos\left(\omega t\right)u\left(t\right)$	$\frac{s}{s^2+\omega^2}$
$\sin\left(\omega t\right)u\left(t\right)$	$\frac{\omega}{s^2+\omega^2}$

Table 1: Several time-domain function and their Laplace transforms

Note the use of $u\left(t\right)$ to ensure that the time-domain signals are zero for $t < 0$. If a text does not include this explicitly, it can be assumed to be the case.

2.2. PROPERTIES OF THE LAPLACE TRANSFORM

As well as the above results for the Laplace transform of a given time-domain function (time signal) the transform also has several important fundamental properties.

Differentiation

What happens if we wish to evaluate the Laplace transform of the derivative of a given signal: $L\left[\frac{df}{dt}\right]$?

From the definition

$$L\left[\frac{df}{dt}\right] = \int_0^\infty \frac{df}{dt} e^{-st} dt$$

Use integration by parts, and let $u = e^{-st}$ and $dv = df$. Then

$$\begin{aligned} L\left[\frac{df}{dt}\right] = \int_0^\infty e^{-st}\frac{df}{dt}dt &= \left[e^{-st} f\left(t\right)\right]_0^\infty - \int_0^\infty \left(-se^{-st}\right) f\left(t\right) dt \\ &= \left[0 - f\left(0\right)\right] + s\int_0^\infty f\left(t\right) e^{-st} dt \qquad (14) \\ &= sF\left(s\right) - f\left(0\right) \end{aligned}$$

Hence, since $L\left[\frac{df}{dt}\right] = sF\left(s\right) - f\left(0\right)$, the Laplace transform has the important property of turning time-domain calculus (the operation of differentiation) into *s*-domain algebra (the operation of multiplying by *s*).

Integration

Now, what if we wish to evaluate the Laplace transform of the integral of a given signal?

$$L\left[\int_0^t f\left(\tau\right) d\tau\right] = \int_0^\infty \left(\int_0^t f\left(\tau\right) d\tau\right) e^{-st} dt$$

Again use integration by parts, this time choosing $u = \int_0^t f\left(\tau\right) d\tau$ and $dv = e^{-st}$, hence giving

$$\begin{aligned} L\left[\int_0^t f\left(\tau\right) d\tau\right] &= \int_0^\infty \left(\int_0^t f\left(\tau\right) d\tau\right) e^{-st} dt \\ &= \left[\left(\int_0^t f\left(\tau\right) d\tau\right)\left(-\frac{1}{s}e^{-st}\right)\right]_0^\infty - \int_0^\infty f\left(t\right)\left(-\frac{1}{s}e^{-st}\right) dt \\ &= \left[0-0\right] + \frac{1}{s}\int_0^\infty f\left(t\right) e^{-st} dt = \frac{1}{s}F\left(s\right) \end{aligned}$$

Hence the time-domain operation of integration is transformed into the simple operation of dividing by *s* in the *s*-domain:

$$L\left[\int_0^t f(\tau)\,d\tau\right] = \frac{1}{s}F(s)$$

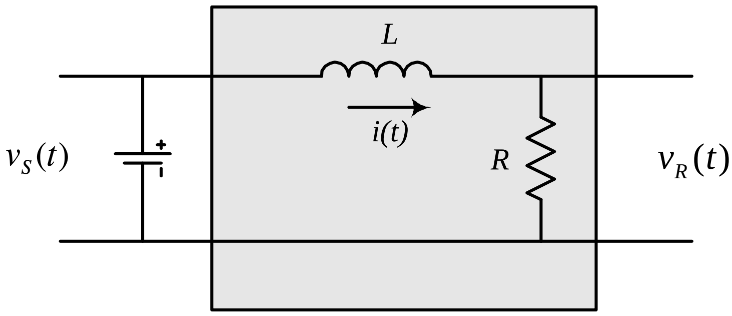
Example 3 : Laplace transform of a unit ramp

If we wish to evaluate $L\left[tu(t)\right]$ then it is simplest just to note that $tu(t)$ is the integral of the unit step function $u(t)$.

Hence, using the integration result, we get:

$$\begin{aligned} L\left[tu(t)\right] &= \frac{1}{s}L\left[u(t)\right] \\ &= \frac{1}{s}\times\frac{1}{s} = \frac{1}{s^2} \end{aligned}$$

Example 4 : LR circuit with Thévenin power source



Consider again the previous example of an LR circuit, with $v_S(t)$ as the driving input, and the resistor voltage as the output.
The input is a linear ramp, $v_S(t) = kt$ for $t \geq 0$ for some constant *k*, and $v_R(0) = 0$.

We already know that the differential equation that describes the circuit is:

$$\frac{L}{R}\frac{dv_R}{dt} + v_R = kt$$

Taking the Laplace transform of the entire equation, term by term, gives:

$$\frac{L}{R}\left[sV_R\left(s\right)-v_R\left(0\right)\right]+V_R\left(s\right)=\frac{k}{s^2}$$

Since $v_R\left(0\right)=0$, rearranging gives:

$$\left[{sL}/{R}+1\right]V_R\left(s\right)=\frac{k}{s^2}$$

$$\Rightarrow \quad V_R\left(s\right)=\frac{k}{s^2\left({sL}/{R}+1\right)}$$

This is the *s*-domain version of the full output (particular solution) in the time domain - it contains the same information as the complementary function, the particular integral and the initial conditions. But, to get the time-domain output, we need to carry out the *inverse Laplace transform* of this expression.

It is possible to work out inverse Laplace transforms using integrals, but fortunately, for engineering purposes, it is normally adequate to use tables of signals, such as those illustrated in Table 1.

Most of the time, the *s*-domain output won't appear in the tables in a recognisable form, but it can usually be broken into a sum of simpler terms using partial fractions.

In this case, for example, write

$$\frac{k}{s^2\left({sL}/{R}+1\right)}=\frac{A}{s^2}+\frac{B}{s}+\frac{C}{{sL}/{R}+1}$$

and evaluate the unknown coefficients to obtain

$$\frac{k}{s^2\left({sL}/{R}+1\right)}=\frac{k}{s^2}-\frac{{kL}/{R}}{s}+\frac{{kL}/{R}}{s+{R}/{L}}$$

so that we can use tables to conclude that

$$v_R\left(t\right)=\underbrace{kt-\frac{kL}{R}}_{\text{STEADY STATE}}+\underbrace{\frac{kL}{R}\exp\left(-{Rt}/{L}\right)}_{\text{TRANSIENT}}\quad \text{for } t\geq 0$$

as before.

The key idea is to use the Laplace transform and its inverse to avoid calculus by changing the problem into a three-stage process:

$$\begin{tikzcd}
\text{D.E.} \arrow[r, "\text{Laplace transform}"] \arrow[d, "\text{calculus}"'] & s\text{-domain equation} \arrow[d, "\text{algebra}"] \\
t\text{-domain solution} & s\text{-domain solution} \arrow[l, "\text{inverse Laplace}"']
\end{tikzcd}$$

The use of tables for both the transform and inverse transform stage reduces the workload very considerably - the main effort is in the algebra involved in:

(1) Rearranging the equation to obtain the solution.

(2) Rearranging the solution in order to use tables (partial fractions).

2.3. PARTIAL FRACTIONS

There are various ways to work out partial fraction expansions but some (such as the cover-up rule) don't always work. The most systematic (and quickest) approach is:

(1) Ensure that the degree of the numerator polynomial $P\left(x\right)$ (degree *m*) is less than that of the denominator polynomial $Q\left(x\right)$ (degree *n*) - if not, first carry out a polynomial division and then consider just the remainder term.

(2) Make sure that the expansion contains all the appropriate terms. For each factor:

$\left(ax+b\right)$ include a term $\dfrac{A}{\left(ax+b\right)}$

$\left(ax+b\right)^k$ include terms $\dfrac{A}{\left(ax+b\right)}+\dfrac{B}{\left(ax+b\right)^2}+\cdots+\dfrac{K}{\left(ax+b\right)^k}$

$\left(ax^2+bx+c\right)$ include a term $\dfrac{Ax+B}{\left(ax^2+bx+c\right)}$

$\left(ax^2+bx+c\right)^k$ include terms $\dfrac{Ax+B}{\left(ax^2+bx+c\right)}+\cdots+\dfrac{Jx+K}{\left(ax^2+bx+c\right)^k}$

where *A*, *B*, *C*,... are numbers. The final expansion will look something like

$$\frac{P(x)}{Q(x)}=\sum_i \frac{A_i}{\text{powers of linear factors}}+\sum_j \frac{B_j x+C_j}{\text{powers of quadratic factors}}$$

For example - simple linear factors:

$$\frac{cx+d}{(x+a)(x+b)}=\frac{A}{(x+a)}+\frac{B}{(x+b)}$$

and one quadratic factor:

$$\frac{cx^2+dx+e}{(x+a)\left(x^2+b\right)}=\frac{A}{(x+a)}+\frac{Bx+C}{\left(x^2+b\right)}$$

or one repeated linear factor:

$$\frac{cx^2+dx+e}{(x+a)(x+b)^2}=\frac{A}{(x+a)}+\frac{B}{(x+b)}+\frac{C}{(x+b)^2}$$

(3) Multiply both sides by $Q(x)$. This will produce an identity between two polynomials, each of degree *n*-1. It is usually worth writing out this step in full.

(4) Obtain relationships between the *n* unknown coefficients on the r.h.s. and the known coefficients of $P(x)$ on the l.h.s.

(5) Solve these relationships for the values of the unknown coefficients.

The third step is straightforward, but the next two can be extremely difficult if not handled carefully. Common approaches are to insert *n* particular, distinct values of *x* in both sides of the equation, or to expand both sides fully into polynomials of order *n*-1 and to compare the coefficients of x^i for $0 \le i \le n-1$. Both of these approaches results in n simultaneous algebraic equations in the n unknown coefficients, which then require further effort to obtain a full solution.

A good approach is to use particular values of x which cause many of the linear factors in the r.h.s. to vanish: choose x to be a root of $Q\left(x\right)$.

This will produce simpler equations for some of the unknown coefficients. The problem is that, in all but the simplest cases, repeated roots and/or quadratic factors will still cause problems: there are not enough independent linear factors to give n equations in the unknown coefficients.

The most useful approach for the quadratic factors is to calculate a complex root of any quadratic factor and set x to be that complex root on both sides of the identity. Some manipulation of complex numbers will be required but, after simplification, it is then possible to compare separately the real and imaginary parts of the equation to obtain two pieces of information about the coefficients at once.

In general it may be necessary to combine a little of each of the above approaches: the aim is to produce a set of relationships between the unknown coefficients that will require as little effort as possible to solve for specific numerical values.

Example 5 :

Expand $\dfrac{s+1}{s\left(s+2\right)}$ in partial fractions.

Write the ratio as $\dfrac{s+1}{s\left(s+2\right)} = \dfrac{A}{s} + \dfrac{B}{\left(s+2\right)}$

Then multiply through by $s\left(s+2\right)$ to get

$$s+1 \equiv A\left(s+2\right) + Bs$$

(and note that this is now an *identity* - valid for *any* value of s).

Let

$$s=0 \quad \Rightarrow \quad 1=2A \quad \Rightarrow \quad A=\frac{1}{2}$$

and $$s=-2 \quad \Rightarrow \quad -1=-2B \quad \Rightarrow \quad B=\frac{1}{2}$$

So that we have

$$\frac{s+1}{s\left(s+2\right)}=\frac{1/2}{s}+\frac{1/2}{\left(s+2\right)}$$

Example 6 :

Expand $\dfrac{s+1}{\left(s+2\right)^2\left(s+3\right)}$ in partial fractions.

Write the ratio as $\dfrac{s+1}{\left(s+2\right)^2\left(s+3\right)}=\dfrac{A}{\left(s+2\right)^2}+\dfrac{B}{\left(s+2\right)}+\dfrac{C}{\left(s+3\right)}$

Then multiply through by $\left(s+2\right)^2\left(s+3\right)$ to get

$$s+1\equiv A\left(s+3\right)+B\left(s+2\right)\left(s+3\right)+C\left(s+2\right)^2$$

Again, this is now an identity, valid for any value of *s*.

Let

$$s=-2 \quad\Rightarrow\quad -1=A \quad\Rightarrow\quad A=-1$$

and $\quad s=-3 \quad\Rightarrow\quad -2=C \quad\Rightarrow\quad C=-2$

Finally, compare the coefficients of s^2:

$$0=B+C \quad\Rightarrow\quad B=-C \quad\Rightarrow\quad B=2$$

Hence,

$$\frac{s+1}{\left(s+2\right)^2\left(s+3\right)}=-\frac{1}{\left(s+2\right)^2}+\frac{2}{\left(s+2\right)}-\frac{2}{\left(s+3\right)}$$

Example 7 :

Expand $\dfrac{s+1}{\left(s^2+4\right)\left(s+3\right)}$ in partial fractions.

Write the ratio as $\dfrac{s+1}{\left(s^2+4\right)\left(s+3\right)}=\dfrac{As+B}{\left(s^2+4\right)}+\dfrac{C}{\left(s+3\right)}$

Then multiply through by $\left(s^2+4\right)\left(s+3\right)$ to get

$$s+1 \equiv (As+B)(s+3)+C(s^2+4)$$

Since this is an identity, valid for any value of *s*, let

$$s=-3 \quad \Rightarrow \quad -2=13C \quad \Rightarrow \quad C=-2/13$$

and
$$\begin{aligned} s=2j \quad \Rightarrow \quad 1+2j &= (B+2jA)(3+2j) \\ &= (3B-4A)+(6A+2B)j \end{aligned}$$

so that equating real and imaginary parts gives:

$$\begin{aligned} -4A+3B &= 1 \\ 6A+2B &= 2 \end{aligned}$$

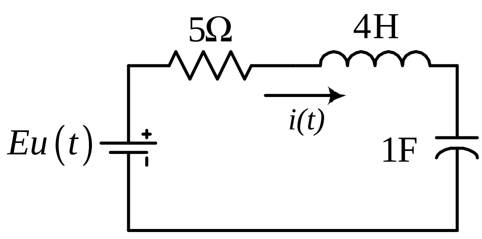
so that

$$A = 2/13 \quad \text{and} \quad B = 7/13$$

Hence
$$\frac{s+1}{(s^2+4)(s+3)} = \frac{1}{13}\left[\frac{2s+7}{(s^2+4)} - \frac{2}{(s+3)}\right]$$

Example 8 :

Consider the circuit



with zero initial conditions, and find $i(t)$.

In this example Kirchhoff's Voltage Law gives

$$4\frac{di}{dt}+5i+\int_0^t i(\tau)\,d\tau + v_C(0) = Eu(t)$$

$v_C(0)=0$, so that the Laplace transform of the equation is

$$4\left[sI\left(s\right)-i\left(0\right)\right]+5I\left(s\right)+\frac{1}{s}I\left(s\right)=E\frac{1}{s}$$

and since $i\left(0\right)=0$ this is

$$\left(4s+5+\frac{1}{s}\right)I\left(s\right)=E\frac{1}{s}$$

$$\Rightarrow I\left(s\right)=\frac{E}{s\left(4s+5+{}^{1}\!/_{s}\right)}=\frac{E}{\left(4s^2+5s+1\right)}$$

$$=\frac{E}{\left(4s+1\right)\left(s+1\right)}=\frac{{}^{E}\!/_{4}}{\left(s+{}^{1}\!/_{4}\right)\left(s+1\right)}$$

$$=\frac{{}^{E}\!/_{3}}{\left(s+{}^{1}\!/_{4}\right)}-\frac{{}^{E}\!/_{3}}{\left(s+1\right)}$$

Hence, using tables, the time-domain output current is:

$$i\left(t\right)=\frac{E}{3}\left(e^{-{}^{t}\!/_{4}}-e^{-t}\right)$$

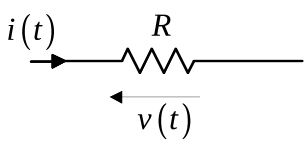
2.4. LAPLACE TRANSFORMS FOR CIRCUIT COMPONENTS

As seen above, it is quite straightforward to write down the differential equations for a circuit and then to use the Laplace transform to switch into a viewpoint where the problem can be solved using only algebra, rather than calculus.

An even quicker approach, however, is to visualise the properties of the entire circuit directly in the *s*-domain.

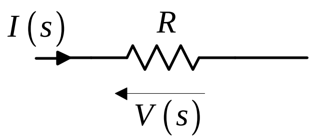
The resistor

In the time domain, the resistor is viewed as



$$v\left(t\right)=Ri\left(t\right)$$

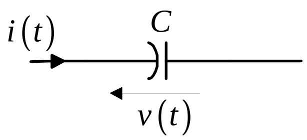
while in the *s*-domain we instead have



$$V(s) = RI(s)$$

The capacitor

In the time domain, the capacitor is viewed as



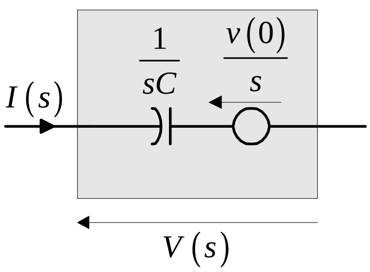
$$i(t) = C\frac{dv(t)}{dt} \quad \text{or} \quad v(t) = \frac{1}{C}\int_0^t i(\tau)\,d\tau + v(0)$$

Laplace transforming the first of these (equivalent) relationships gives the *s*-domain viewpoint:

$$I(s) = C\left[sV(s) - v(0)\right]$$

$$\Rightarrow V(s) = \frac{1}{sC}I(s) + \frac{v(0)}{s}$$

and the model for a charged capacitor can be shown as:

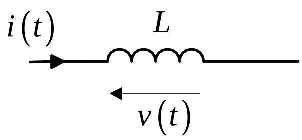


where the energy associated with any initial charge on the capacitor is modelled via an idealised voltage source.

We say that the *Laplace impedance* of a capacitor is $1/_{sC}$.

The inductor

In the time domain, the inductor is viewed as

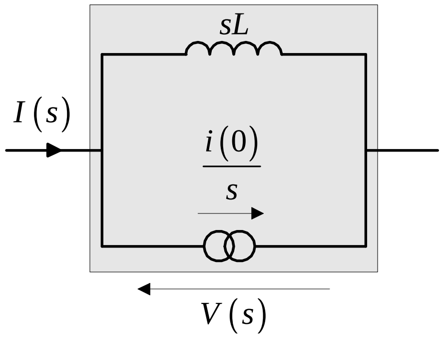


$$v(t) = L\frac{di(t)}{dt} \quad \text{or} \quad i(t) = \frac{1}{L}\int_0^t v(\tau)\,d\tau + i(0)$$

Laplace transforming gives:

$$I(s) = \frac{1}{sL}V(s) + \frac{i(0)}{s}$$

and the *s*-domain inductor model is:

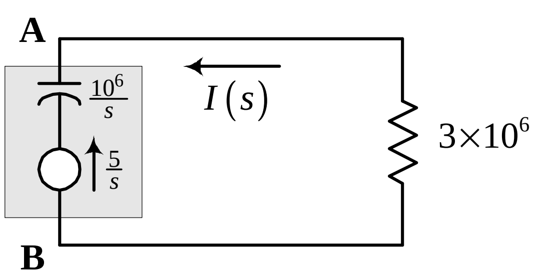


where the energy associated with any initial flux through the inductor is modelled via an idealised current source.

sL is the Laplace impedance of an inductor.

Example 9 :

A capacitor of $1\mu\text{F}$ is charged to 5V and connected to a $3\text{M}\Omega$ resistor at $t = 0$. What current flows?



For this circuit the total impedance is $\frac{10^6}{s} + 3\times 10^6$ and so

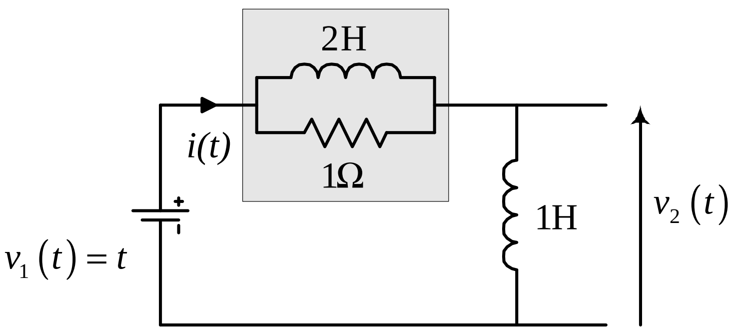
$$I(s) = \frac{-\,^5\!/_s}{\frac{10^6}{s} + 3\times 10^6} = \frac{-5\times 10^{-6}}{3s+1} = \frac{-\,^5\!/_3 \times 10^{-6}}{s + \,^1\!/_3}$$

Hence the inverse Laplace transform (using tables) gives:

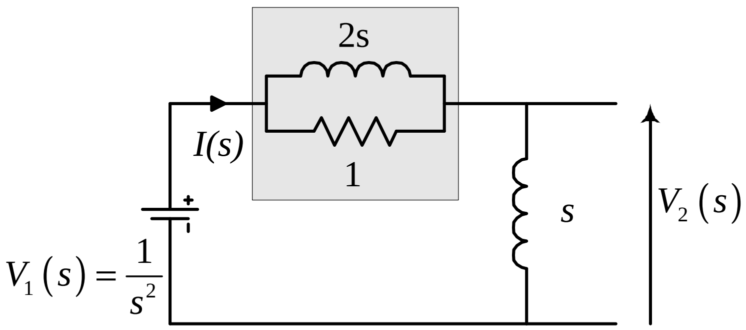
$$i(t) = -\frac{5}{3} e^{-t/3} \ \mu\text{A} \,.$$

Example 10 :

In the circuit below, what is the output voltage $v_2(t)$ if there are zero initial conditions?



In Laplace form, we have:



The resistor and the Laplace impedance of the inductor can be combined using the normal result for parallel impedances:

$$\left(Z = {Z_1 Z_2} \big/ {\left(Z_1 + Z_2 \right)} \right)$$

Hence $$I\left(s\right) = \frac{V_1\left(s\right)}{Z\left(s\right)} = \frac{\frac{1}{s^2}}{\left(s + \frac{2s}{2s+1}\right)}$$

and so $$V_2\left(s\right) = sI\left(s\right) = \frac{\frac{1}{s}}{\left(s + \frac{2s}{2s+1}\right)} = \frac{2s+1}{2s^3 + 3s^2} = \frac{s + 1/2}{s^2\left(s + 3/2\right)}$$

Partial fractions gives:

$$V_2\left(s\right) = \frac{1/3}{s^2} + \frac{4/9}{s} + \frac{-4/9}{\left(s + 3/2\right)}$$

Finally, taking the inverse Laplace transforms using tables gives:

$$v_2\left(t\right) = \underbrace{1/3\, t + 4/9}_{\text{FORCED RESPONSE}} \underbrace{-4/9\, e^{-3t/2}}_{\text{TRANSIENT RESPONSE}} \qquad t \geq 0$$

or $$v_2\left(t\right) = \underbrace{1/3\, tu\left(t\right) + 4/9\, u\left(t\right)}_{\text{FORCED RESPONSE}} \underbrace{-4/9\, e^{-3t/2} u\left(t\right)}_{\text{TRANSIENT RESPONSE}}$$

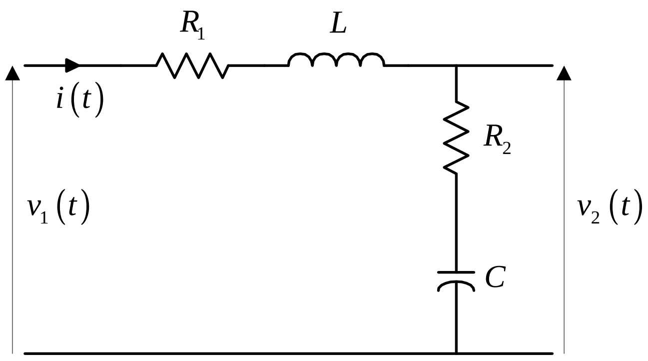
2.5. LUMPED COMPONENTS AND NETWORK FUNCTIONS

In the examples above, the resistors, inductors and capacitors are idealised models of physical characteristics that, in practice, may be distributed throughout the circuit. In general, the resistance, inductance and capacitance may be spread continuously and unevenly along the wires but, for simplicity, their cumulative effects are instead represented by isolated elements placed at particular points in the circuit - these are called *lumped components*.

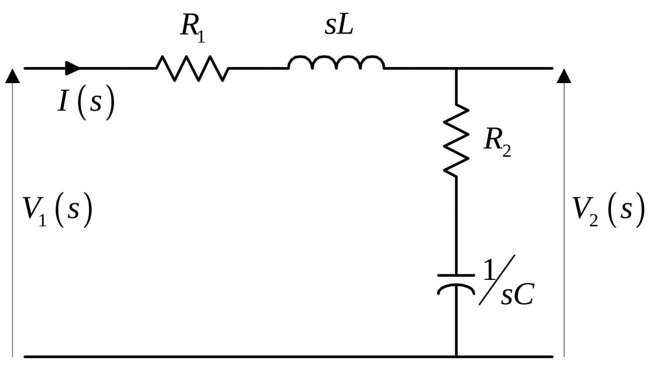
The lumped component approach provides a good approximation in a wide variety of circuit applications and allows the straightforward modelling of large, multi-loop network topologies. When combined with the Laplace techniques illustrated above, however, it is possible to go even further, and to represent the relationships between signals at different points in the circuits in the form of *network functions* - *s*-space expressions that contain all of the information about not only the values of the various circuit components, but also their relative positions within the network.

Example 11 :

Consider the circuit below, with output $v_2(t)$, zero initial conditions and an input signal $v_1(t)$.



The Laplace version of the circuit is



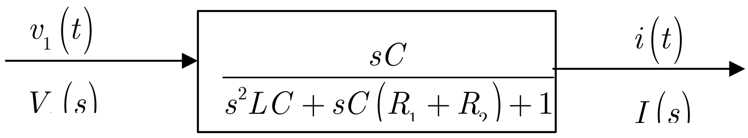
The loop impedance is $R_1 + sL + R_2 + {}^1\!/_{sC}$ and hence

$$I(s) = \frac{1}{R_1 + sL + R_2 + {}^1\!/_{sC}} V_1(s)$$

$$\Rightarrow I(s) = \left(\frac{sC}{s^2LC + sC(R_1 + R_2) + 1} \right) V_1(s)$$

This describes a network function, (or *transfer function* or *input-*

output relationship) between the driving voltage and the loop current:



$\frac{V_1}{I}(s) = \frac{s^2LC + sC(R_1 + R_2) + 1}{sC}$ is another network function,

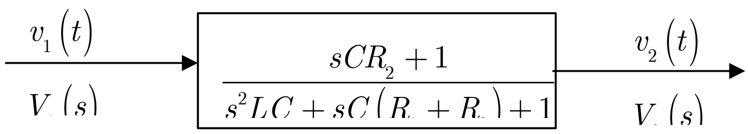
called the driving point input impedance (since voltage and current are measured at the same place).

Likewise,

$$V_2(s) = \frac{R_2 + 1/sC}{R_1 + sL + R_2 + 1/sC} V_1(s)$$

$$\Rightarrow V_2(s) = \left(\frac{sCR_2 + 1}{s^2LC + sC(R_1 + R_2) + 1}\right) V_1(s)$$

describes a transfer function (input-output relationship) between the driving voltage and the output voltage:



This is called the *voltage transfer function*.

$\frac{V_2(s)}{I(s)} = R_2 + 1/sC = \frac{sCR_2 + 1}{sC}$ is called the *transfer impedance*.

Note that all of the above network functions for these lumped component circuit networks are rational functions in *s* with real coefficients. In general, network functions will have the form:

$$H(s) = \frac{a_n s^n + a_{n-1} s^{n-1} + a_{n-2} s^{n-2} + \cdots + a_1 s + a_0}{b_m s^m + b_{m-1} s^{m-1} + b_{m-2} s^{m-2} + \cdots + b_1 s + b_0}$$

where the a_i and b_j are real numbers. Such a rational function can always be written in the form:

$$H(s) = G\frac{(s-z_1)(s-z_2)(s-z_3)\cdots(s-z_{n-1})(s-z_n)}{(s-p_1)(s-p_2)(s-p_3)\cdots(s-p_{m-1})(s-p_m)}$$

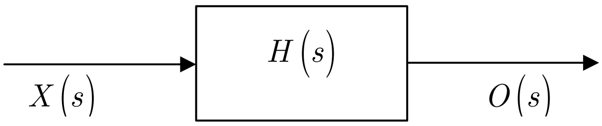
where $G = \frac{a_n}{b_m}$ is the *gain constant* of the system.

The $(s-z_i)$ are the zero factors and the z_i are the system zeroes.

The $(s-p_i)$ are the pole factors and the p_i are the system poles.

The poles and zeroes are the roots of the denominator and numerator polynomials, respectively.

Note that the network function is *completely* characterised by a knowledge of the various poles, zeroes and the system gain.



Here, $$O(s) = H(s)X(s)$$

where $X(s)$ is the Laplace input, $O(s)$ is the Laplace output, and $H(s)$ is the *system function* that relates the output to the input..

Particular named versions arise in circuit network theory:

$$V(s) = Z(s)I(s) \qquad \text{and} \qquad I(s) = Y(s)V(s)$$

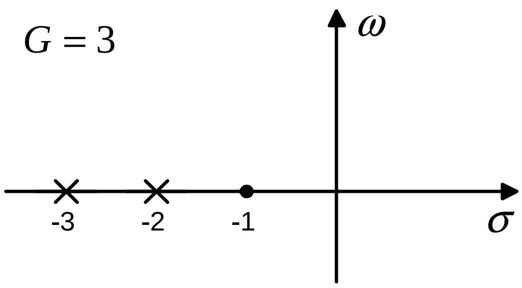
where $Z(s)$ is the impedance function and $Y(s) = {}^{1}\!/_{Z(s)}$ is the admittance function.

Example 12 :

A lumped component system has the following network function relating two signals:

$$H(s) = 3\frac{(s+1)}{(s+2)(s+3)}$$

In *s*-space we would draw:



where the complex variable $s = \sigma + j\omega$.

2.6. *FREQUENCY RESPONSE OF A SYSTEM*

To establish the frequency response (sinusoidal response) of a system $H(s)$, apply an input function $x(t) = \sin(\omega t)$ and consider the output:

$$O(s) = H(s)X(s)$$

and if $x(t) = \sin(\omega t)$ then $X(s) = \dfrac{\omega}{s^2 + \omega^2}$

If $H(s)$ has the form

$$H(s) = G\frac{(s - z_1)(s - z_2)(s - z_3)\cdots(s - z_{n-1})(s - z_n)}{(s - p_1)(s - p_2)(s - p_3)\cdots(s - p_{m-1})(s - p_m)}$$

$$= G\frac{\prod_{i=1}^{n}(s - z_i)}{\prod_{j=1}^{m}(s - p_j)}$$

then it is possible to carry out a partial fractions expansion of the output

$$O(s) = G\frac{\prod_{i=1}^{n}(s - z_i)}{\prod_{j=1}^{m}(s - p_j)}\left(\frac{\omega}{s^2 + \omega^2}\right)$$

$$= \frac{As + B}{s^2 + \omega^2} + \sum_{j=1}^{m}\frac{C_j}{s - p_j}$$

Hence, taking the normal steps for partial fractions,

$$G\prod_{i=1}^{n}\left(s-z_i\right)\omega=\left(As+B\right)\prod_{j=1}^{m}\left(s-p_j\right)+\left(s^2+\omega^2\right)\sum_{k=1}^{m}\frac{C_k\prod_{j=1}^{m}\left(s-p_j\right)}{s-p_k}$$

Setting $s=j\omega$ gives

$$G\prod_{i=1}^{n}\left(j\omega-z_i\right)\omega=\left(Aj\omega+B\right)\prod_{j=1}^{m}\left(j\omega-p_j\right)$$

so that $$Aj\omega+B=G\frac{\prod_{i=1}^{n}\left(j\omega-z_i\right)}{\prod_{j=1}^{m}\left(j\omega-p_j\right)}\omega=\omega H\left(j\omega\right)$$

Likewise setting $s=-j\omega$ gives

$$-Aj\omega+B=G\frac{\prod_{i=1}^{n}\left(-j\omega-z_i\right)}{\prod_{j=1}^{m}\left(-j\omega-p_j\right)}\omega=\omega H\left(-j\omega\right)$$

If we write $$H\left(j\omega\right)=\left|H\left(j\omega\right)\right|\exp\left(j\phi\left(j\omega\right)\right)$$

then also $$H\left(-j\omega\right)=\left|H\left(j\omega\right)\right|\exp\left(-j\phi\left(j\omega\right)\right)$$

So that $$Aj\omega+B=\omega\left|H\left(j\omega\right)\right|\exp\left(j\phi\left(j\omega\right)\right)$$

and $$-Aj\omega+B=\omega\left|H\left(j\omega\right)\right|\exp\left(-j\phi\left(j\omega\right)\right)$$

Subtracting gives

$$2Aj\omega=\omega\left|H\left(j\omega\right)\right|\left[\exp\left(j\phi\left(j\omega\right)\right)-\exp\left(-j\phi\left(j\omega\right)\right)\right]$$

$$\Rightarrow A=\left|H\left(j\omega\right)\right|\frac{\left[\exp\left(j\phi\left(j\omega\right)\right)-\exp\left(-j\phi\left(j\omega\right)\right)\right]}{2j}$$

$$\Rightarrow A=\left|H\left(j\omega\right)\right|\sin\left(\phi\left(j\omega\right)\right)$$

and adding gives

$$\begin{aligned}
2B &= \omega\left|H\left(j\omega\right)\right|\left[\exp\left(j\phi\left(j\omega\right)\right)+\exp\left(-j\phi\left(j\omega\right)\right)\right]\\
\Rightarrow B &= \omega\left|H\left(j\omega\right)\right|\frac{\left[\exp\left(j\phi\left(j\omega\right)\right)+\exp\left(-j\phi\left(j\omega\right)\right)\right]}{2}\\
\Rightarrow B &= \omega\left|H\left(j\omega\right)\right|\cos\left(\phi\left(j\omega\right)\right)
\end{aligned}$$

Hence

$$\begin{aligned}
O\left(s\right) &= G\frac{\prod_{i=1}^{n}\left(s-z_i\right)}{\prod_{j=1}^{m}\left(s-p_j\right)}\left(\frac{\omega}{s^2+\omega^2}\right)\\
&= \frac{\left|H\left(j\omega\right)\right|\sin\left(\phi\left(j\omega\right)\right)s+\left|H\left(j\omega\right)\right|\cos\left(\phi\left(j\omega\right)\right)\omega}{s^2+\omega^2}+\sum_{j=1}^{m}\frac{C_j}{s-p_j}
\end{aligned}$$

The first part of this expression represents the driven output, while the latter terms within the sum are all transient components, due entirely to the poles of the transfer function alone.

Taking the inverse Laplace transform by tables, we get

$$\begin{aligned}
o\left(t\right) = &\left|H\left(j\omega\right)\right|\sin\left(\phi\left(j\omega\right)\right)\cos\left(\omega t\right)\\
&+\left|H\left(j\omega\right)\right|\cos\left(\phi\left(j\omega\right)\right)\sin\left(\omega t\right)+r\left(t\right)
\end{aligned}$$

where $r\left(t\right)$ is the sum of all of the transient components.

Hence, the steady-state output is

$$o\left(t\right)=\left|H\left(j\omega\right)\right|\sin\left(\omega t+\phi\left(j\omega\right)\right)$$

It follows that, given a transfer function $H\left(s\right)$ and a need to find the steady-state sinusoidal response, simply set $s=j\omega$ and calculate the magnitude $\left|H\left(j\omega\right)\right|$ and phase angle $\phi\left(j\omega\right)$ of $H\left(j\omega\right)$.

Example 13 :

When a 1 amp current source $i\left(t\right)$ is turned on to a two terminal circuit, a voltage $v\left(t\right)$ is developed, where

$$v\left(t\right)=\frac{1}{2}e^{-t}+\frac{1}{2}e^{-3t}$$

Determine

(a) The impedance of the circuit

(b) The steady-state current which would flow if a sinusoidal voltage source of peak value 1 volt and frequency 1 rad/s were applied.

Part (a)

$$\text{If } i(t) = u(t) \text{ then } I(s) = \frac{1}{s}.$$

$$\text{Also, if } v(t) = \frac{1}{2}e^{-t} + \frac{1}{2}e^{-3t} \text{ then}$$

$$V(s) = \frac{1/2}{s+1} + \frac{1/2}{s+3} = \frac{s+2}{(s+1)(s+3)}$$

Hence the circuit impedance is

$$Z(s) = \frac{V(s)}{I(s)} = \frac{s(s+2)}{(s+1)(s+3)}$$

Part (b)

We now have $I(s) = \dfrac{1}{Z(s)}V(s) = \dfrac{(s+1)(s+3)}{s(s+2)}$

The general frequency response function is obtained by setting $s = j\omega$ in the transfer function $H(s)$ $(= \dfrac{1}{Z(s)}$ in this case).

$$H(j\omega) = \frac{(1+j\omega)(3+j\omega)}{j\omega(2+j\omega)}$$

But we are asked to consider the particular case of an input of $\sin(t)$:

$$H(j) = \frac{(1+j)(3+j)}{j(2+j)} = \frac{-j(1+j)(3+j)(2-j)}{5}$$

$$= \frac{(1-j)(7-j)}{5} = \frac{6-8j}{5} = 1.2 - 1.6j = 2\angle -0.93$$

Hence the steady-state current has a peak value of 2 amps and lags (negative phase angle) the voltage by a phase of 0.93 radians.

2.7. PLOTTING POLES AND ZEROES

In general, transfer functions will have the form

$$H(s) = G\frac{(s-z_1)(s-z_2)(s-z_3)\cdots(s-z_{n-1})(s-z_n)}{(s-p_1)(s-p_2)(s-p_3)\cdots(s-p_{m-1})(s-p_m)}$$

For example, if we have the particular case of

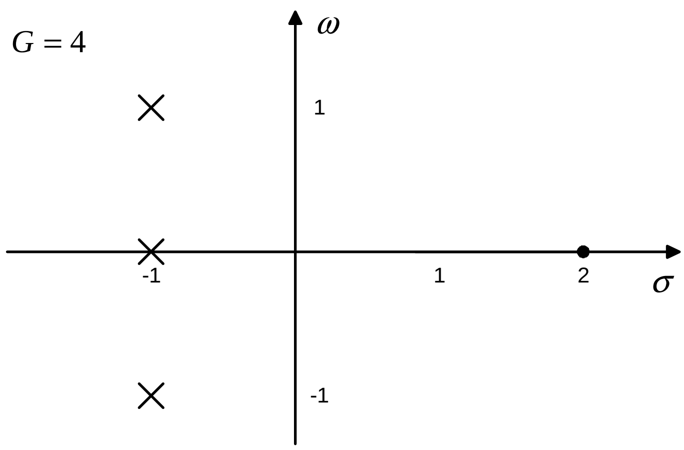
$$H(s) = \frac{4(s-2)}{(s+1)(s^2+2s+2)}$$

this can be written as

$$H(s) = 4\frac{(s-2)}{(s+1)\left[(s+1)^2+1\right]} = 4\frac{(s-2)}{(s+1)(s+1-j)(s+1+j)}$$

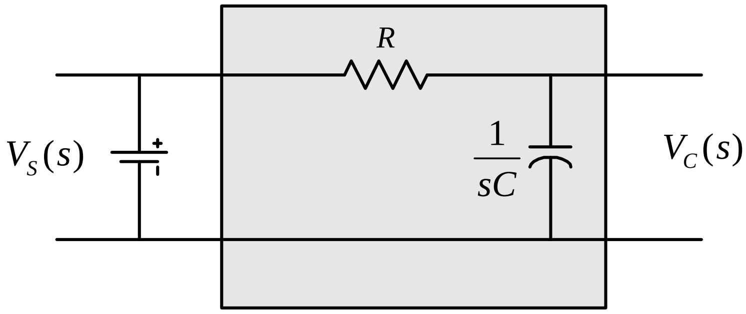
so that we have a system with one zero (at $s=2$) and three poles (one real pole at $s=-1$ and a complex conjugate pair at $s=-1\pm j$).

This gives rise to a pole-zero plot of the form



NB - all polynomials with real coefficients have roots which are either real or appear as complex conjugate pairs.

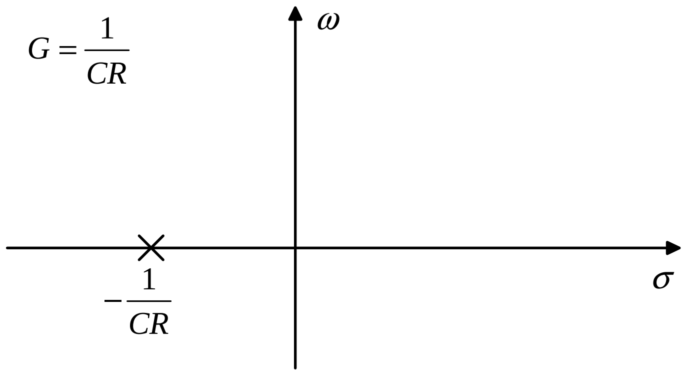
Example 14 : RC circuit



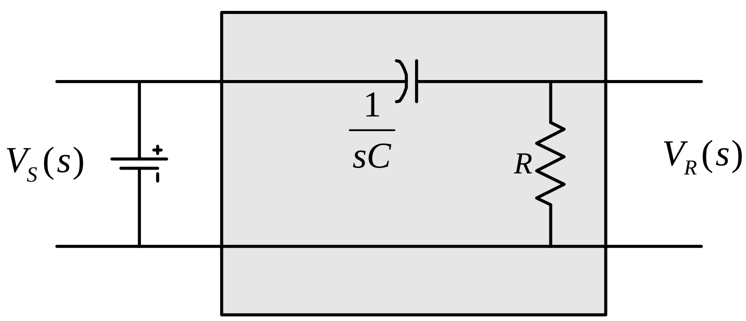
Here,

$$\frac{V_C(s)}{V_S(s)} = H(s) = \frac{1/sC}{R + 1/sC} = \frac{1}{sCR+1} = \frac{1}{CR}\frac{1}{\left(s + 1/CR\right)}$$

so that the pole-zero plot is:

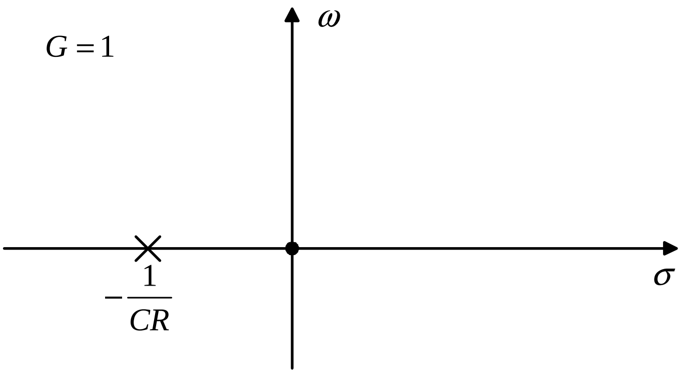


Example 15 : RC circuit



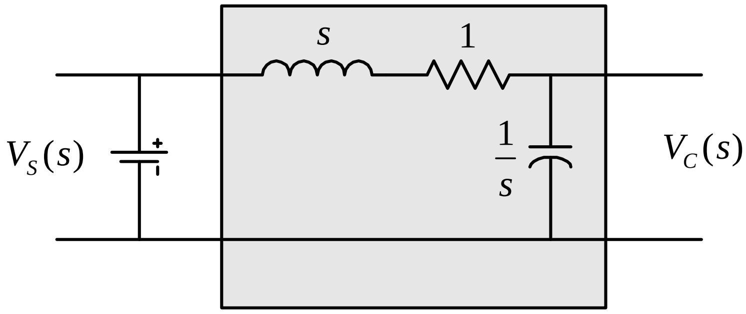
Here $$H\left(s\right)=\frac{R}{R+{}^{1}\!/\!{}_{sC}}=\frac{sCR}{sCR+1}=\frac{s}{s+{}^{1}\!/\!{}_{CR}}$$

so that in this case there is also a zero at the origin:



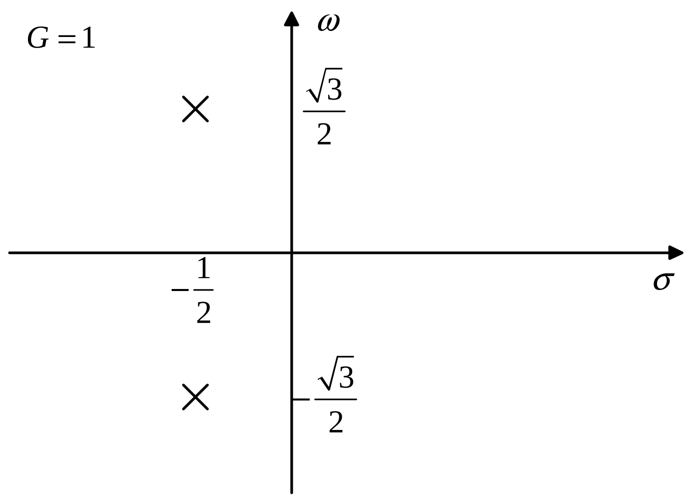
Example 16 : Series RLC circuit

Consider a series RLC circuit with component values of $L=1\mathrm{H}$, $R=1\Omega$ and $C=1\mathrm{F}$.



$$\begin{aligned}\frac{V_C\left(s\right)}{V_S\left(s\right)}&=H\left(s\right)\\&=\frac{{}^{1}\!/\!{}_{s}}{s+1+{}^{1}\!/\!{}_{s}}=\frac{1}{s^2+s+1}\\&=\frac{1}{\left(s+\frac{1}{2}+j\,\frac{\sqrt{3}}{2}\right)\left(s+\frac{1}{2}-j\,\frac{\sqrt{3}}{2}\right)}\end{aligned}$$

so that the pole-zero plot shows a pair of complex conjugate poles:



2.8. IMPULSE RESPONSE AND SYSTEM STABILITY

In general, all passive circuits will have their poles in the LHP (left-hand plane), but this need not be the case for all systems.

Consider the general input-output relationship in Laplace form

$$O(s) = H(s)X(s)$$

The input $x(t)$ could take many forms:

- if a step function, we would be able to examine the "step response" of the system;
- if a sinusoidal signal of variable frequency, we would be able to record the "frequency response" (Bode plots);
- and if a Dirac delta function is applied as the input, we obtain the "impulse response".

Recall that $L[\delta(t)] = 1$. Then, if it were possible to apply a perfect delta function as the input (it is *not* possible in practice), the Laplace transform of the output would be:

$$O(s) = H(s) \times 1 = H(s)$$

so that in the time domain

$$o(t) = L^{-1}[H(s)] = h(t)$$

$h(t)$ is referred to as the *impulse response* of this general linear system, and we can say that *the Laplace transform of the impulse response* $h(t)$ *is the system transfer function* $H(s)$.

The impulse response is a crucial concept in engineering systems, even though the idealised impulse cannot be obtained in practice.

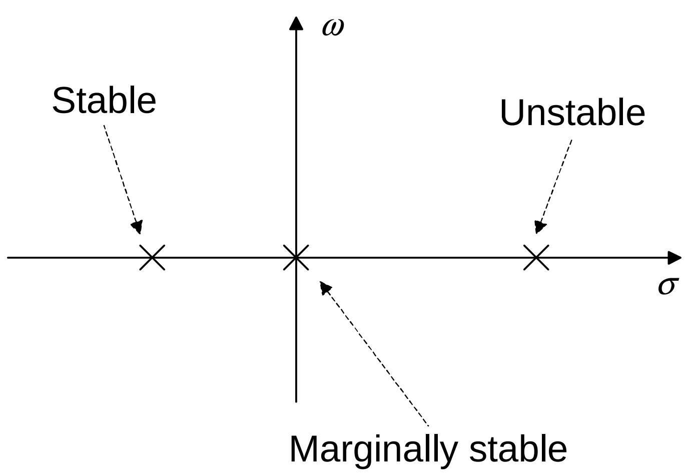
Next, consider a general form of $H(s)$, where the *m* poles, p_j, can re real or appear as complex conjugate pairs, and expand it in partial fractions:

$$H(s) = G\frac{\prod_{i=1}^{n}(s - z_i)}{\prod_{j=1}^{m}(s - p_j)} = \sum_{j=1}^{m}\frac{k_j}{s - p_j}$$

For each real pole, the impulse response will have a component of the form $k_j \exp(p_j t)$. Hence, for negative values of p_j, these components will be decaying exponentials, for positive values they will be increasing exponentials, and for $p_j = 0$, they will contribute a constant value to the output.

In terms of the *s*-plane this means that poles that appear on the negative real axis contribute stable (decaying) terms to the output, while poles on the positive real axis indicate unstable (rising) solutions.

A single pole at the origin is described as being marginally stable.



The only other types of poles that can appear within $H(s)$ are complex conjugate pairs, say $p = \sigma + j\omega$ and $p^* = \sigma - j\omega$. In the partial fractions expansion these terms will pair up in the form

$$\frac{k}{s-\sigma-j\omega} + \frac{k^*}{s-\sigma+j\omega}$$

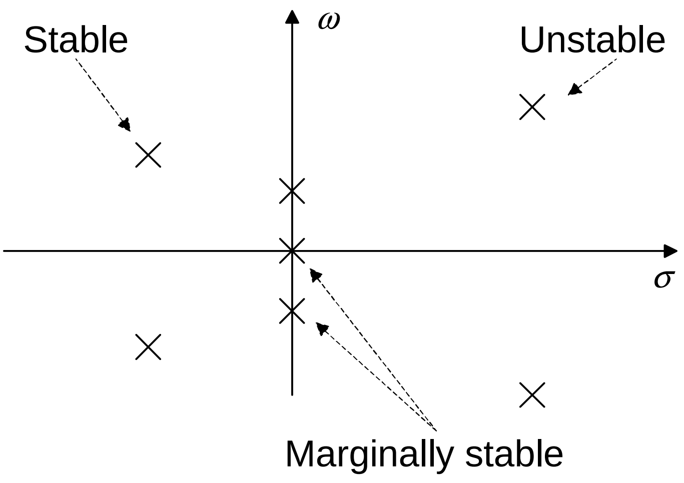
Let $k = |k|\exp(j\phi)$. Then $k^* = |k|\exp(-j\phi)$ and the Laplace transform of the expression above is

$$\begin{aligned} ke^{(\sigma+j\omega)t} + k^* e^{(\sigma-j\omega)t} &= |k|\, e^{j\phi} e^{(\sigma+j\omega)t} + |k|\, e^{-j\phi} e^{(\sigma-j\omega)t} \\ &= |k|\, e^{\sigma t}\left(e^{j(\omega t+\phi)} + e^{-j(\omega t+\phi)}\right) = 2|k|\, e^{\sigma t}\cos(\omega t+\phi) \end{aligned}$$

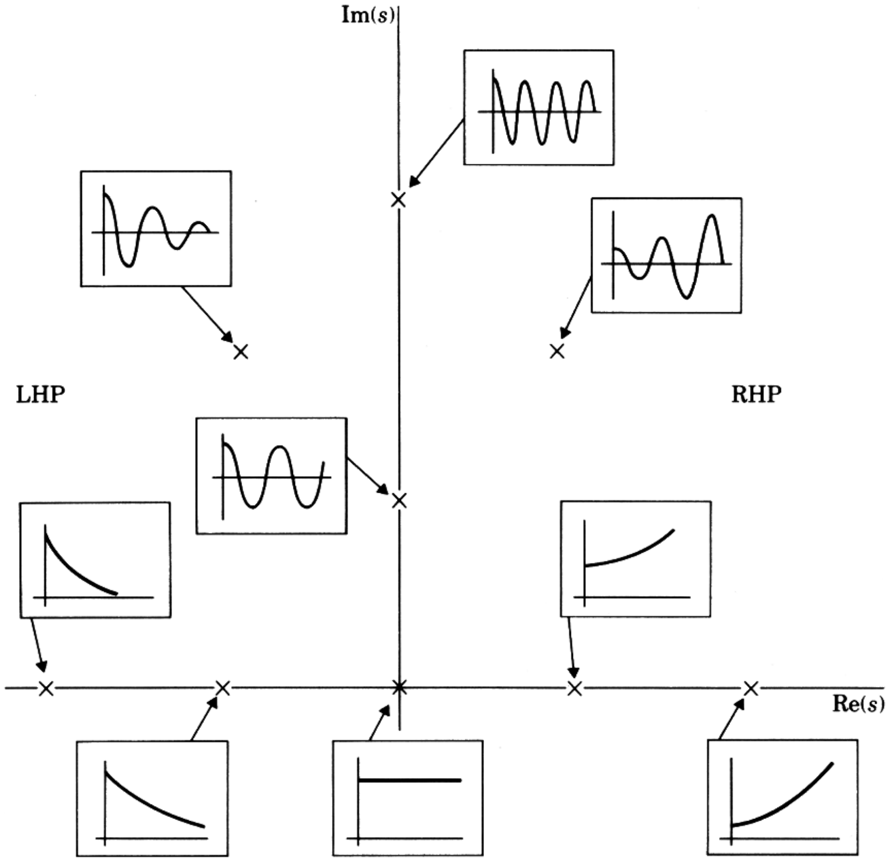
For negative values of σ such contributions to the output have the form of decaying sinusoidal waves - and are hence stable.

For positive values of σ, the sinusoids will be increasing in magnitude and are hence unstable.

When $\sigma = 0$, the contribution represents a sinusoidal wave of constant magnitude and is described as being marginally stable.



Overall, for stability, a system must have *all* of the poles of $H(s)$ - real and complex - in the LHP (Left-Hand Plane). Such as system is described as being *asymptotically stable*, since the outputs will settle towards zero asymptotically as time increases.



Poles that are actually on the imaginary axis are called marginally stable, provided that they are not repeated (i.e. "simple" poles). If repeated, such roots are unstable, since they involve powers of time.

For example,

$$L^{-1}\left[\frac{k}{\left(s-j\omega\right)^{r}}\right]=\frac{k}{\left(r-1\right)!}t^{r-1}e^{j\omega t}$$

which is unstable if $r>1$.

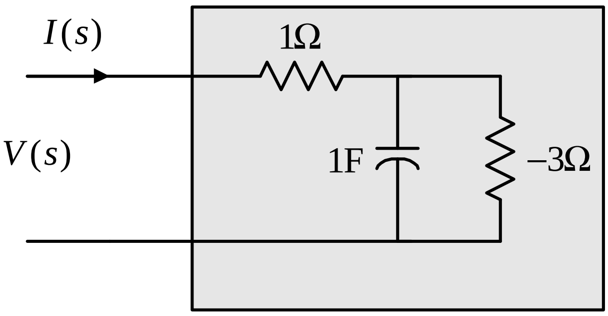
In summary, for a linear time-invariant system:

- if all poles have negative real parts, the system is asymptotically stable.
- if any poles with a zero real part are simple, and all other poles have negative real parts, the system is marginally stable.
- if any poles with a zero real part are repeated, the system is unstable.
- if any poles have positive real parts, the system is unstable.

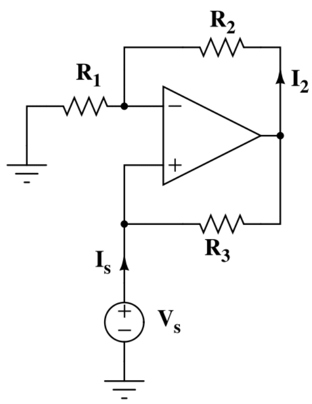
Example 17 : Stability of impedance and admittance functions

In general, the stability of a system is determined only by the poles of the transfer function - the zeroes play no part. But this can depend upon the viewpoint being taken.

Consider the circuit



Note that the negative resistor can be created using a Negative Impedance Converter (NIC).

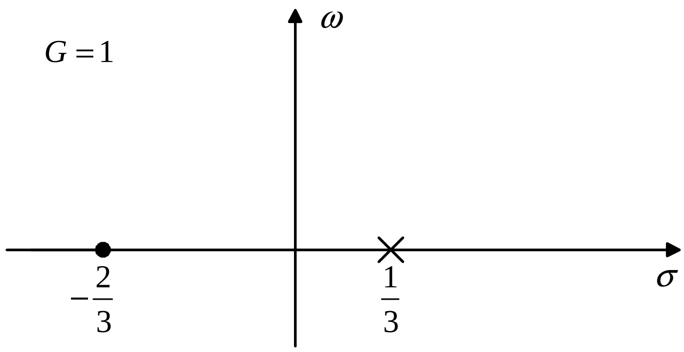


Here, for example, the input resistance is given by $R_{in} = -R_3 \dfrac{R_1}{R_2}$.

The Laplace impedance of the circuit is

$$\frac{V(s)}{I(s)} = Z(s) = 1 + \frac{\frac{1}{s}(-3)}{\frac{1}{s}+(-3)} = 1 + \frac{3}{3s-1} = \frac{3s+2}{3s-1} = \frac{s+{}^{2}\!/_{3}}{s-{}^{1}\!/_{3}}$$

The pole-zero plot is



In short, when viewed in the form

$$V(s) = Z(s)I(s)$$

this represents an unstable system - whatever current source is applied, the output voltage will contain an element that increases towards infinity over time.

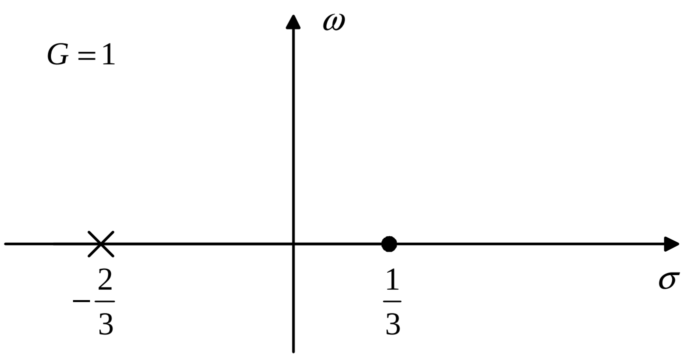
However, when we take the view that there is a voltage source and that the current is the output, we get

$$I(s) = \frac{1}{Z(s)}V(s) = Y(s)V(s)$$

where, in this case,

$$Y(s) = \frac{s - \frac{1}{3}}{s + \frac{2}{3}}$$

which now represents a stable system with a pole-zero plot of the form:

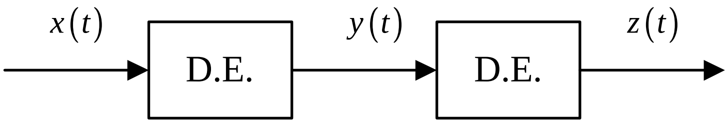


The circuit is said to be *open circuit unstable* but *short-circuit stable*.

2.9. *TRANSFER FUNCTIONS AND SERIES SYSTEMS*

One of the key benefits of working with Laplace variables is the fact that calculus operations are transformed into algebraic ones - and this is particularly useful when working with series systems.

For example, in the time domain, the input and output signals are generally related by differential equations:



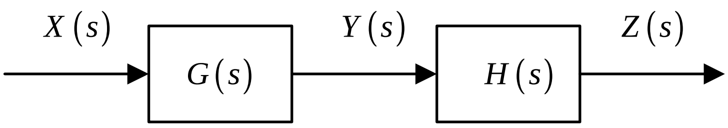
Here, $x(t)$ is the input to some dynamic system whose output is $y(t)$, and $y(t)$ is in turn the input to a second dynamic system whose output is $z(t)$.

Solving this series system in the time domain would require either:

(a) completely solving the first differential equation, including initial conditions, to obtain the output $y(t)$ for all values of time, and then separately using this output to allow the solution of the next equation for the final output $z(t)$.

(b) combining the two equations to obtain one (higher-order) differential equation with input $x(t)$ and output $z(t)$ - effectively, eliminating $y(t)$.

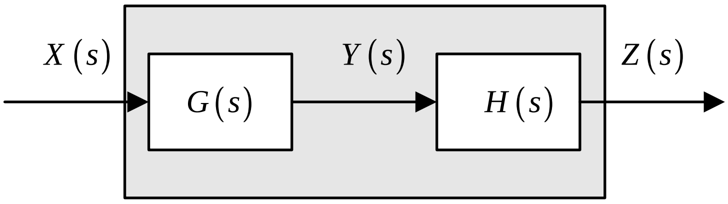
In the Laplace domain, the situation is simplified considerably. Each differential equation is represented by a transfer function, and these can simply be combined algebraically (by multiplication).



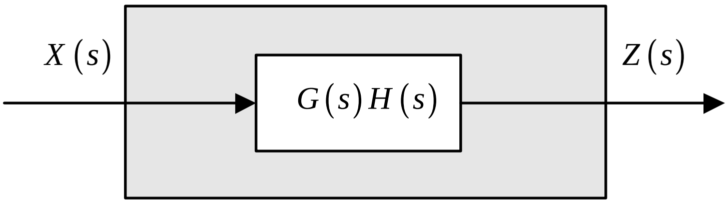
Here,

$$Y(s) = G(s)X(s) \quad \text{and} \quad Z(s) = H(s)Y(s)$$

so that $$Z(s) = H(s)G(s)X(s) = \left[G(s)H(s)\right]X(s)$$



is the same as



3. PROPERTIES OF LAPLACE TRANSFORMS

So far, we have seen the Laplace transforms of a few simple signals, and some basic properties - that integration in the time-domain corresponds to multiplication by $\frac{1}{s}$ in the s-domain, while differentiation corresponds to multiplying by s. But there are a variety of other powerful results that are useful for handling engineering signals and systems.

3.1. THE FIRST SHIFTING THEOREM

If $L\left[f\left(t\right)\right]=F\left(s\right)$, then $L\left[e^{at}f\left(t\right)\right]=F\left(s-a\right)$.

Proof

$$\begin{aligned} L\left[e^{at}f\left(t\right)\right] &= \int_0^\infty e^{at}f\left(t\right)e^{-st}dt \\ &= \int_0^\infty f\left(t\right)e^{-\left(s-a\right)t}dt = F\left(s-a\right) \qquad \text{QED} \end{aligned}$$

Example 18

Find $L\left[e^{at}\sin\left(\omega t\right)\right]$.

Note that $$L\left[\sin\left(\omega t\right)\right]=\frac{\omega}{s^2+\omega^2}$$

and use the result $L\left[e^{at}f\left(t\right)\right]=F\left(s-a\right)$

so that $L\left[e^{at}\sin\left(\omega t\right)\right]=\dfrac{\omega}{\left(s-a\right)^2+\omega^2}=\dfrac{\omega}{s^2-2as+\left(a^2+\omega^2\right)}$

3.2. THE SECOND SHIFTING THEOREM

If $L\left[f\left(t\right)\right]=F\left(s\right)$, then $L\left[u\left(t-T\right)f\left(t-T\right)\right]=e^{-sT}F\left(s\right)$.

Note that the signal $u\left(t-T\right)f\left(t-T\right)$ is merely a delayed version of the signal $f\left(t\right)$ - it is zero until $t=T$, at which point the signal "switches on".

Proof

$$\begin{aligned}L\left[u\left(t-T\right)f\left(t-T\right)\right]&=\int_0^\infty u\left(t-T\right)f\left(t-T\right)e^{-st}dt\\&=\int_T^\infty f\left(t-T\right)e^{-st}dt\end{aligned}$$

Make a change of variables - let $u=t-T$. Then

$$dt=du \quad \text{and} \quad t=T \quad \Leftrightarrow \quad u=0$$

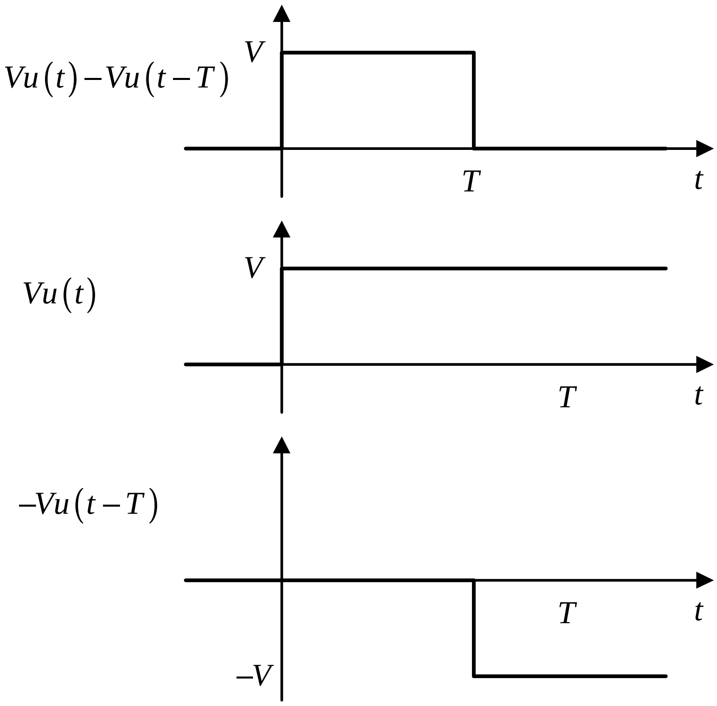
So that

$$\begin{aligned}L\left[u\left(t-T\right)f\left(t-T\right)\right]&=\int_T^\infty f\left(t-T\right)e^{-st}dt\\&=\int_0^\infty f\left(u\right)e^{-s\left(u+T\right)}du\\&=e^{-sT}\int_0^\infty f\left(u\right)e^{-su}du=e^{-sT}F\left(s\right) \qquad \text{QED}\end{aligned}$$

Example 19

Calculate the Laplace transform of a rectangular voltage pulse of amplitude *V* and duration *T*.

The pulse can be created by adding the step function signal $Vu\left(t\right)$ to the delayed (and negative) step function $-Vu\left(t-T\right)$:



Hence $L\left[Vu\left(t\right)-Vu\left(t-T\right)\right]=V\frac{1}{s}-V\frac{e^{-sT}}{s}=\frac{V}{s}\left(1-e^{-sT}\right)$

Example 20

Calculate the Laplace transform of a signal $f\left(t\right)$ that has the form of a single half cycle of a sine wave of period *T*.

In this case, it is easiest to create the sinusoidal pulse by adding the sine wave to a time-shifted version of itself:

$$f\left(t\right)=u\left(t\right)\sin\left(\omega t\right)+u\left(t-\tfrac{T}{2}\right)\sin\left(\omega\left(t-\tfrac{T}{2}\right)\right)$$

these two signals will cancel perfectly after the time $t=\frac{T}{2}$, and so the required Laplace transform is

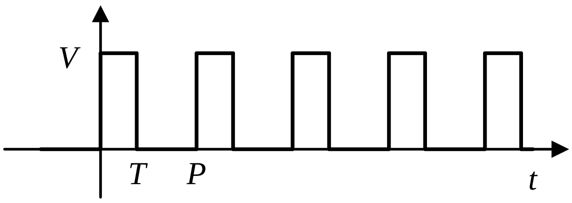
$$F\left(s\right)=\frac{\omega}{s^2+\omega^2}+e^{-sT/2}\frac{\omega}{s^2+\omega^2}=\frac{\omega}{s^2+\omega^2}\left(1+e^{-sT/2}\right)$$

Note also that, by definition, $\omega T=2\pi$ so that the result can be written in either of the equivalent forms

$$F\left(s\right)=\frac{\omega}{s^2+\omega^2}\left(1+e^{-s\pi/\omega}\right)=\frac{2\pi/T}{s^2+\left(2\pi/T\right)^2}\left(1+e^{-sT/2}\right)$$

Example 21

Calculate the Laplace transform of a rectangular pulse train of strength *V*, period *P* and pulse width *T*



Here, a rectangular pulse, say $f\left(t\right)$, occurs at time $t = 0$ and then again at time $t = P$, $t = 2P$, etc. The full time signal is hence

$$f_P\left(t\right) = f\left(t\right) + f\left(t-P\right) + f\left(t-2P\right) + f\left(t-3P\right) + \ldots = \sum_{n=0}^{\infty} f\left(t-nP\right)$$

- an infinite series of rectangular pulses.

The first pulse can be constructed as the sum of two step functions - one of size *V*, turned on at time *t*=0, and one of size -*V*, turned on at time *t*=*T*.

The next pulse is similarly constructed of two step functions at time *T* and *T*+*P*, and so on:

$$f_P\left(t\right) = \sum_{n=0}^{\infty} f\left(t-nP\right) = \sum_{n=0}^{\infty}\left[Vu\left(t-nP\right) - Vu\left(t-T-nP\right)\right]$$
$$= V\sum_{n=0}^{\infty}\left[u\left(t-nP\right) - u\left(t-T-nP\right)\right]$$

Hence

$$F_P\left(s\right) = V\sum_{n=0}^{\infty}\left[\frac{1}{s}e^{-snP} - \frac{1}{s}e^{-snP-sT}\right]$$
$$= \frac{V}{s}\left[1-e^{-sT}\right]\sum_{n=0}^{\infty} e^{-snP}$$
$$= \frac{V}{s}\left[1-e^{-sT}\right]\left[1+e^{-sP}+e^{-2sP}+e^{-3sP}+\cdots\right]$$

The last portion of this is a geometric series with initial term 1 and ratio e^{-sP} - assuming convergence, we get

$$1+e^{-sP}+e^{-2sP}+e^{-3sP}+\cdots=\frac{1}{1-e^{-sP}}$$

so that for the Laplace transform of a rectangular pulse train of period *P* and pulse width *T* is

$$F_P(s)=\frac{V}{s}\left[\frac{1-e^{-sT}}{1-e^{-sP}}\right]$$

Example 22

Calculate the Laplace transform of a square wave.

For the particular case of a square wave, we would have $V=1$ and $T=P/2$. Hence

$$F_P(s)=\frac{1}{s}\frac{\left(1-e^{-sP/2}\right)}{\left(1-e^{-sP}\right)}=\frac{1}{s}\frac{\left(1-e^{-sP/2}\right)}{\left(1-e^{-sP/2}\right)\left(1+e^{-sP/2}\right)}=\frac{1}{s\left(1+e^{-sP/2}\right)}$$

Example 23

Calculate the Laplace transform of a signal of pulses of the general form $f(t)$ and period *P*.

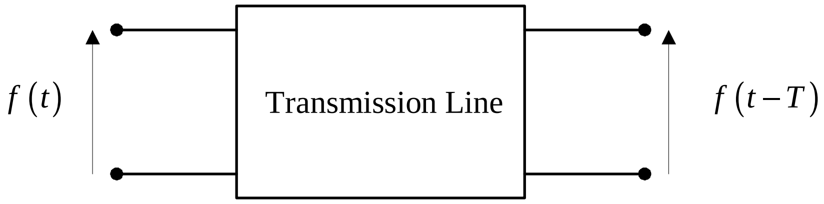
As before, the time signal has the form

$$f_P(t)=\sum_{n=0}^{\infty}f(t-nP)$$

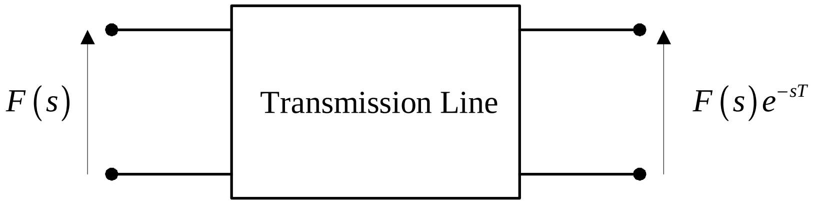
$$\Rightarrow F_P(s)=\sum_{n=0}^{\infty}F(s)e^{-snP}=F(s)\sum_{n=0}^{\infty}e^{-snP}=\frac{F(s)}{1-e^{-sP}}$$

Example 24

An ideal transmission line merely delays a signal, rather than changing its frequency content. What is its frequency response?



In Laplace terms this is:



So that the system transfer function is

$$G\left(s\right)=\frac{F\left(s\right)e^{-sT}}{F\left(s\right)}=e^{-sT}$$

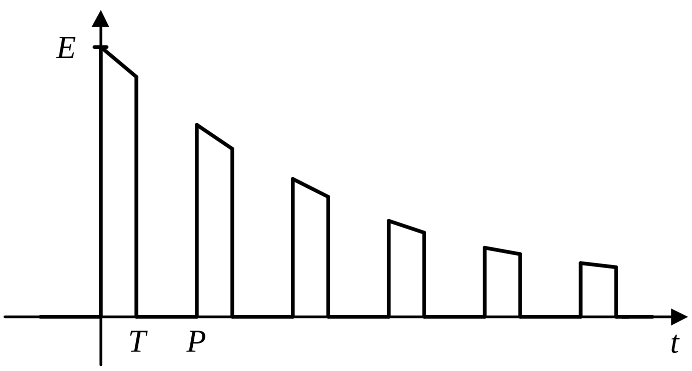
The frequency response of this function is obtained by setting $s=j\omega$:

$$\Rightarrow\left|G\left(j\omega\right)\right|=\left|e^{-j\omega T}\right|=\left|\cos\left(\omega T\right)-j\sin\left(\omega T\right)\right|=1$$

$$\Rightarrow\arg\left(G\left(j\omega\right)\right)=-\omega T$$

Example 25

Obtain the Laplace transform of an exponentially decaying pulse train:



In this case the signal is the product of a rectangular pulse train, $f_P\left(t\right)$ of strength *E*, period *P* and pulse width *T*, with an exponential *envelope* of the form $\exp(-\alpha t)$.

Recall that the first shifting theorem is

If $L\left[f(t)\right]=F(s)$, then $L\left[e^{at}f(t)\right]=F(s-a)$

and that for the rectangular pulse train

$$F_P(s)=\frac{E}{s}\left[\frac{1-e^{-sT}}{1-e^{-sP}}\right]$$

Hence the required Laplace function is

$$\frac{E}{(s+\alpha)}\left[\frac{1-e^{-(s+\alpha)T}}{1-e^{-(s+\alpha)P}}\right]$$

3.3. <u>*THE INITIAL VALUE THEOREM*</u>

If $L\left[f(t)\right]=F(s)$, then $f(0)=\lim_{s\to\infty}\left[sF(s)\right]$.

Proof

We know that $L\left[\frac{df}{dt}\right]=sF(s)-f(0)$

$$\Rightarrow sF(s)=f(0)+\int_0^\infty \frac{df}{dt}(t)e^{-st}dt$$

Note that we have been so far been assuming that e^{-st} will tend to zero as *t* increases - if this is the case, then it must also tend to zero as s increases. Hence

$$\begin{aligned}\lim_{s\to\infty}\left[sF(s)\right]&=\lim_{s\to\infty}\left[f(0)+\int_0^\infty \frac{df}{dt}(t)e^{-st}dt\right]\\&=\lim_{s\to\infty}\left[f(0)\right]+\lim_{s\to\infty}\left[\int_0^\infty \frac{df}{dt}(t)e^{-st}dt\right]\\&=\lim_{s\to\infty}\left[f(0)\right]+\int_0^\infty \frac{df}{dt}(t)\lim_{s\to\infty}\left[e^{-st}\right]dt\\&=\lim_{s\to\infty}\left[f(0)\right]+0=\lim_{s\to\infty}\left[f(0)\right]\qquad\text{QED}\end{aligned}$$

NOTE: There is a potential convergence problem with the argument above - many texts will quote the differentiation result as

$$L\left[\frac{df}{dt}\right] = sF(s) - f(0_+)$$

where $f(0_+)$ means the limiting value as we approach the origin from the positive side - strictly, the above proof is not valid at $t = 0$ and can give a misleading result if there is, for example, a delta function at the origin.

The more correct form of the result is that

$$f(0_+) = \lim_{s\to\infty}\left[sF(s)\right].$$

provided that the transfer function $F(s)$ is *strictly proper* - which means that the degree of the numerator polynomial is lower than that of the denominator (there are more poles than zeroes).

In practice, for engineering purposes, this is not a problem unless there is an impulse (and hence a discontinuity) at the origin.

Example 26

What is the initial value of the function $f(t)$ if $F(s) = \dfrac{E}{s+\alpha}$?

In this case $sF(s) = \dfrac{sE}{s+\alpha} = \dfrac{E}{1+\dfrac{\alpha}{s}}$

which tends to E as $s \to \infty$. Hence $f(0) = E$.

(It's easy to check that, in fact, $f(t) = Ee^{-\alpha t}$).

Example 27

A system has a transfer function $G(s) = \dfrac{2s+3}{s^2+2s+4}$ Establish the initial value of the impulse response and the initial value of the gradient of the impulse response.

Recall that the Laplace transform of the impulse response *is* the system transfer function - hence the initial value of the impulse response is

$$g(0) = \lim_{s\to\infty}\left[sG(s)\right] = \lim_{s\to\infty}\left[\frac{2s^2+3s}{s^2+2s+4}\right] = 2\,.$$

For the gradient, first note that in general, since

$$L\left[\frac{df}{dt}\right] = sF(s) - f(0)$$

$$L\left[\frac{d^2f}{dt^2}\right] = L\left[\frac{d}{dt}\left(\frac{df}{dt}\right)\right] = s\left(sF(s) - f(0)\right) - \frac{df}{dt}(0)$$

$$= s^2F(s) - sf(0) - \frac{df}{dt}(0)$$

and hence, as before, $\frac{df}{dt}(0) = \lim_{s\to\infty}\left[s^2F(s) - sf(0)\right]$

In this particular case

$$\frac{dg}{dt}(0) = \lim_{s\to\infty}\left[s^2\left(\frac{2s+3}{s^2+2s+4}\right) - 2s\right]$$

$$= \lim_{s\to\infty}\left[\frac{2s^3+3s^2}{s^2+2s+4} - \frac{2s^3+4s^2+8s}{s^2+2s+4}\right] = \lim_{s\to\infty}\left[\frac{-s^2-8s}{s^2+2s+4}\right] = -1$$

Hence $g(0) = 2$ and $g'(0) = -1$.

3.4. THE FINAL VALUE THEOREM

If $L\left[f(t)\right] = F(s)$, then $\lim_{t\to\infty}\left[f(t)\right] = \lim_{s\to 0}\left[sF(s)\right]$.

Proof

We know that $L\left[\frac{df}{dt}\right] = sF(s) - f(0)$

$$\Rightarrow sF(s) = f(0) + \int_0^\infty \frac{df}{dt}(t)\,e^{-st}dt$$

Let $s \to 0$:

$$\Rightarrow \lim_{s\to 0}\left[sF\left(s\right)-f\left(0\right)\right]=\lim_{s\to 0}\left[\int_0^\infty \frac{df}{dt}\left(t\right)e^{-st}dt\right]$$

$$=\int_0^\infty \frac{df}{dt}\left(t\right)\lim_{s\to 0}\left[e^{-st}\right]dt$$

$$=\int_0^\infty \frac{df}{dt}\left(t\right)dt=\left[f\left(t\right)\right]_0^\infty$$

$$\Rightarrow \lim_{s\to 0}\left[sF\left(s\right)\right]-f\left(0\right)=f\left(\infty\right)-f\left(0\right)$$

$$\Rightarrow f\left(\infty\right)=\lim_{s\to 0}\left[sF\left(s\right)\right] \qquad \text{QED}$$

NOTE: As with the initial value theorem, there are circumstances where the result can be misleading if the formula is used naively. For a meaningful answer it is necessary that both:

- All of the poles of the transfer function $F\left(s\right)$ must have negative real parts.
- Also, $F\left(s\right)$ must not have more than one pole at the origin.

The first requirement guarantees that all exponential and exponentially-weighted sinusoidal components within the signal will decrease with time, while the second excludes signals which include ramp, quadratic or other higher-power components.

For example, a factor of $\left(s^2+9\right)$ in the denominator would correspond to a purely sinusoidal output - and hence the concept of a "final value" would be meaningless. Likewise a factor $\left(s-2\right)$ would correspond to a signal containing a rising exponential term and s^{n+1} factors would correspond to time signals proportional to t^n if $n>0$. In such cases, the signal diverges and there is again no "final value".

Example 28

If it exists, what is the final value of the output if a step function is input to a system with transfer function $H\left(s\right)=\dfrac{2s+3}{s^2+2s+4}$?

The Laplace transform of the output is

$$O\left(s\right)=\left[\frac{2s+3}{s^2+2s+4}\right]\frac{1}{s}$$

There is only a single power of s in the denominator, and the other poles are $s = -1 \pm j\sqrt{3}$. It is hence valid to apply the final value theorem to obtain,

$$o(\infty) = \lim_{s \to 0}\left[sO(s)\right] = \lim_{s \to 0}\left[s\left[\frac{2s+3}{s^2+2s+4}\right]\frac{1}{s}\right]$$

$$= \lim_{s \to 0}\left[\frac{2s+3}{s^2+2s+4}\right] = \frac{3}{4}$$

3.5. CONVOLUTION

The convolution integral is defined as

$$y(t) = h(t) * x(t) = h * x(t) = \int_0^t h(t-\tau)\,x(\tau)\,d\tau$$

Physically, this represents the driven output of a system with impulse response function $h(t)$ and input $x(t)$ - it is the time-domain version of an input-output relationship. In practice, it also corresponds to situations involving blurring, smoothing, averaging or reverberation, for example.

In the Laplace domain, the process of convolution reduces to simple multiplication:

$$L\left[\int_0^t h(t-\tau)\,x(\tau)\,d\tau\right] = L\left[H(s)\right]L\left[X(s)\right]$$

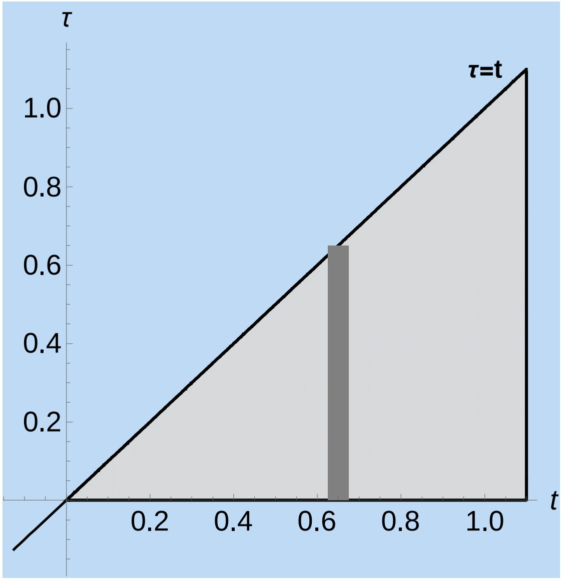
or $$Y(s) = H(s)\,X(s)$$

Proof:

Taking the Laplace transform gives:

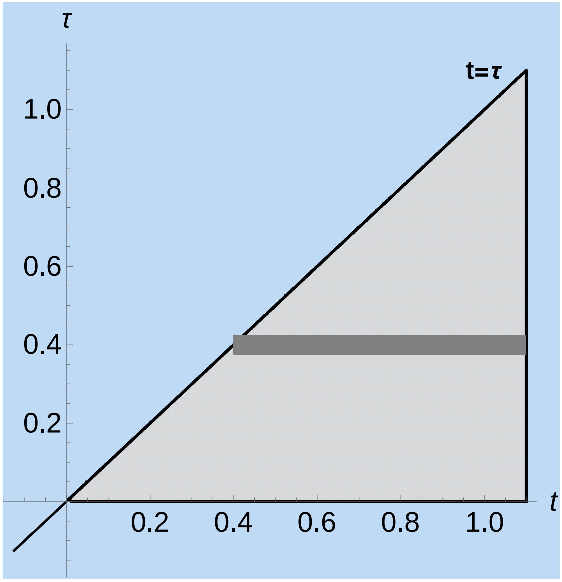
$$Y(s) = \int_0^\infty \left[\int_0^t h(t-\tau)\,x(\tau)\,d\tau\right]e^{-st}dt$$

In this formulation, the τ integration is carried out first (vertically), from the horizontal axis $\tau = 0$ to the line $\tau = t$, defining the infinite triangular region illustrated below.



τ integration followed by t integration.

Reversing the order of integration means that the same region requires first integrating with respect to t (horizontally), from the line $t = \tau$ to $t = \infty$.



t integration followed by τ integration.

Hence

$$\begin{aligned} Y(s) &= \int_0^\infty \left[\int_0^t h(t-\tau)\, x(\tau)\, d\tau \right] e^{-st} dt \\ &= \int_0^\infty \left[\int_\tau^\infty h(t-\tau)\, e^{-st} dt \right] x(\tau) d\tau \end{aligned}$$

since the $x(\tau)$ term can be moved outside of the t integral.

Next, introduce an additional $e^{s\tau}$ term and a compensating $e^{-s\tau}$ term:

$$
\begin{aligned}
Y(s) &= \int_0^\infty \left[\int_\tau^\infty h(t-\tau)e^{-s(t-\tau)}dt\right]x(\tau)e^{-s\tau}d\tau \\
&= \int_0^\infty \left[\int_0^\infty h(v)e^{-sv}dv\right]x(\tau)e^{-s\tau}d\tau
\end{aligned}
$$

where we have first made a change of variables $v = t - \tau$. Hence:

$$
\begin{aligned}
Y(s) &= \int_0^\infty H(s)x(\tau)e^{-s\tau}d\tau \\
&= H(s)\int_0^\infty x(\tau)e^{-s\tau}d\tau = H(s)X(s)
\end{aligned}
$$

which is the standard input-output relationship in Laplace form.

In short, the time-domain integral operation of convolution corresponds to the algebraic operation of multiplication in the *s*-domain.

NB - the reverse result also applies - multiplication (modulation) in the time-domain corresponds to convolution in the *s*-domain.

3.6. SECOND ORDER SYSTEMS AND THE S-PLANE

The transfer function for a general second order system is often written in the form

$$
H(s) = \frac{s-z}{s^2 + 2\zeta\omega_0 s + \omega_0^{\,2}}
$$

where

- *z* is the system zero;
- ζ (zeta) is the *damping ratio* of the system;
- and ω_0 is the *natural frequency* of the system.

The system poles are at

$$
-\zeta\omega_0 \pm \omega_0\sqrt{\zeta^2 - 1}
$$

corresponding to three different forms of time-domain output.

1) If $\zeta > 1$ the system is *overdamped* - there are two distinct real poles, and the output has the form:

$$A_1 \exp\left(\left[-\zeta + \sqrt{\zeta^2 - 1}\right]\omega_0 t\right) + A_2 \exp\left(\left[-\zeta - \sqrt{\zeta^2 - 1}\right]\omega_0 t\right)$$

2) If $\zeta = 1$ the system is *critically damped* - there are two repeated real poles, and the output has the form:

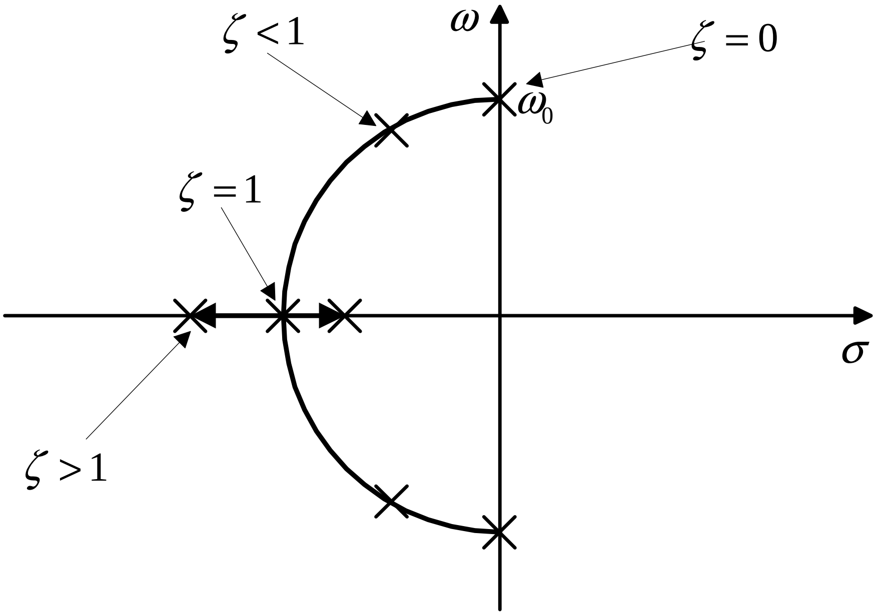
$$A_1 \exp\left(-\omega_0 t\right) + A_2 t \exp\left(-\omega_0 t\right)$$

3) If $\zeta < 1$ the system is *underdamped* - there are a pair of complex conjugate poles and the output has the form:

$$A \exp\left(-\zeta\omega_0 t\right) \sin\left(\sqrt{1-\zeta^2}\,\omega_0 t + \phi\right)$$

As a special case of this last behaviour, if $\zeta = 0$ the system is *undamped* - there are a pair of imaginary conjugate poles and the output has the form $A \sin\left(\omega_0 t + \phi\right)$

In the *s*-plane, these solutions can be drawn as a pole-zero plot:

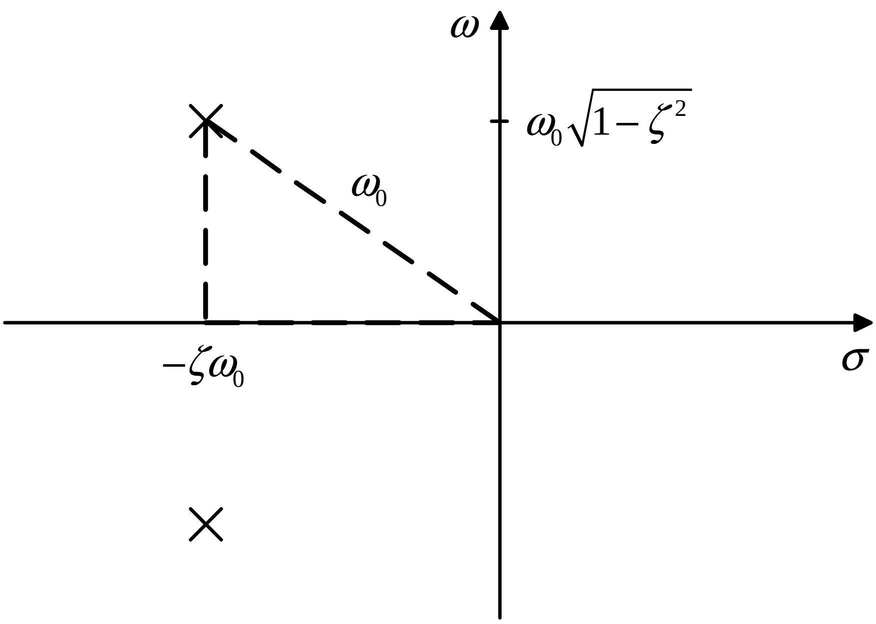


Note that since the two poles are at $-\zeta\omega_0 \pm \omega_0\sqrt{\zeta^2 - 1}$, for underdamped ($\zeta < 1$) systems in particular we can confirm that the poles form a complex conjugate pair at:

$$-\zeta\omega_0 \pm j\omega_0\sqrt{1-\zeta^2}$$

in the *s*-plane.

This gives another important physical insight into the importance of the positions of poles within the *s*-plane:



so that the natural (undamped) frequency corresponds to the distance from the origin, and the damping ratio influences the balance between the real and imaginary parts of the pole.

3.7. POLE-ZERO CANCELLATION

Consider a system with transfer function $H(s) = \frac{5}{s^2 - 1}$ and the input signal $x(t) = \cos(2t) - \frac{1}{2}\sin(2t)$.

Here $$X(s) = \frac{s}{s^2+4} - \frac{1}{2}\frac{2}{s^2+4} = \frac{s-1}{s^2+4}$$

In Laplace space, the output function is

$$Y(s) = H(s)X(s) = \left(\frac{5}{s^2-1}\right)\frac{s-1}{s^2+4} = \frac{5}{(s+1)(s^2+4)}$$

This is an example of pole-zero cancellation, where one of the system poles $s = 1$ appears to be cancelled by a corresponding term in the numerator of the Laplace transform of the system input - the time-domain output would be

$$\begin{aligned} y(t) &= L^{-1}\left[\frac{5}{(s+1)(s^2+4)}\right] \\ &= L^{-1}\left[\frac{1}{(s+1)} + \frac{-s+1}{(s^2+4)}\right] \\ &= L^{-1}\left[\frac{1}{(s+1)} - \frac{s}{(s^2+4)} + \frac{1}{2}\frac{2}{(s^2+4)}\right] \\ &= \exp(-t) - \cos(2t) + \frac{1}{2}\sin(2t) \end{aligned}$$

So that there is no trace of the (unstable) $\exp(t)$ signal associated with the $s = 1$ pole.

Such pole-zero cancellation is impractical in practice - due to issues such as component uncertainties, manufacturing limitations, environmental factors and/or signal noise.

If the zero and pole are reasonably close to each other then, for a while at least, the effect of the (almost cancelled) - pole may not be obvious, but its effect will eventually become significant and, in the case of a pole with a positive real part, the system will still eventually become unstable.

3.8. SUMMARY OF USEFUL TRANSFORMS AND RESULTS

Time-domain	*s*-domain
$\delta(t)$	1
$u(t)$	$\frac{1}{s}$
$tu(t)$	$\frac{1}{s^2}$
$t^n u(t)$	$\frac{n!}{s^{n+1}}$
$e^{at}u(t)$	$\frac{1}{s-a}$
$\cos(\omega t)u(t)$	$\frac{s}{s^2+\omega^2}$
$\sin(\omega t)u(t)$	$\frac{\omega}{s^2+\omega^2}$
$e^{at}\cos(\omega t)u(t)$	$\frac{s-a}{(s-a)^2+\omega^2}$
$e^{at}\sin(\omega t)u(t)$	$\frac{\omega}{(s-a)^2+\omega^2}$
$\frac{df}{dt}$	$sF(s)-f(0)$
$\int_0^t f(\tau)\,d\tau$	$\frac{1}{s}F(s)$
$tf(t)$	$-\frac{dF}{ds}(s)$
Linearity: $af(t)+bg(t)$	$aF(s)+bG(s)$
First shifting theorem: $e^{at}f(t)$	$F(s-a)$
Second shifting theorem: $u(t-T)f(t-T)$	$e^{-sT}F(s)$
Convolution: $y(t)=\int_0^t h(t-\tau)u(\tau)\,d\tau$	$Y(s)=H(s)U(s)$
Initial value theorem: $f(0)=\lim_{s\to\infty}[sF(s)]$	
Final value theorem: $\lim_{t\to\infty}[f(t)]=\lim_{s\to 0}[sF(s)]$	
Laplace impedance of a resistor: R $\quad V(s)=RI(s)$	
Laplace impedance of a capacitor: $\frac{1}{sC}$ $\quad V(s)=\frac{1}{sC}I(s)+\frac{v(0)}{s}$	
Laplace impedance of an inductor: sL $\quad I(s)=\frac{1}{sL}V(s)+\frac{i(0)}{s}$	